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Experimental characterisation and constitutive modelling of the intra-ply tensile and shear properties of unidirectional fibre reinforced thermoplastics (UD FRTPs) under solid-state stamp forming conditions

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ABSTRACT

To enable the success of solid-state stamp forming of unidirectional fibre reinforced thermoplastics (UD FRTPs), it is essential to accurately characterise and model the material deformation under desired conditions. This paper comprehensively investigates the intra-ply tensile and shear properties of unidirectional carbon fibre reinforced polyamide 6 (UD CF/PA6), which is a type of commonly used UD FRTP. To accomplish this, tensile and V-Notched Rail (VNR) shear tests are conducted for characterising the intra-ply transverse tensile and longitudinal shear properties, respectively. The temperature effects (180 – 220 ℃, at 0.01 /s for the transverse tensile deformation and at 0.04 /s for longitudinal shear deformation) and strain-rate effects (0.001 – 0.25 /s for transverse tensile deformation and 0.004 – 0.4 /s for the longitudinal shear deformation, both are at 200 ℃) are studied. It is found that temperature has significant effects on the intra-ply deformation properties, while the strain-rate effects are marginal. This paper also proposes a new physically based constitutive model considering all the deformable constituents, i.e., the polymer constituent reinforced by fibres (PrF) and the polymer-fibre interface (P-F). This model not only shows good prediction of the thermomechanical properties of UD CF/PA6 under intra-ply deformations, but also gives insights into the deformation mechanisms. The new physically based constitutive model is successfully embedded into Finite Element Analysis (FEA) software and validated through accurate prediction of intra-ply deformation of a CF/PA6 specimen under bias-extension. The methodologies and model developed here offer an effective tool for predicting the intra-ply deformation behaviours and guiding the solid-state stamp forming process of UD FRTPs.

1. Introduction

Climate change has driven intensive studies on lightweight materials and their applications in transportation industries to improve operational efficiency and achieve the net-zero target. Of various lightweight materials, fibre reinforced thermoplastics (FRTPs), especially unidirectional carbon fibre reinforced thermoplastics (UD FRTPs), are among the most promising solutions due to their many attractive properties, such as the ability to embed a higher carbon fibre volume fraction compared to woven fibre reinforced polymers (FRPs) [1], superior strength-to-weight ratio, high chemical resistance, good impact resistance and recyclability. To enable their applications in vehicles, a cost-effective manufacturing method that is suitable for mass production is demanded, and the solid-state stamp forming method (i.e., stamp form the FRTPs at temperatures below the melting temperature (T_m) of the matrices) has demonstrated significant potential [2] owing to the short cycle time and energy-saving potential. However, there are still challenges in solid-state stamp forming complex-shaped UD FRTPs due to a lack of understanding of the deformation behaviours. Therefore, for the successful solid-state stamp forming of UD FRTPs, it is essential to accurately characterise the associated deformation mechanisms and to use a physically based constitutive model for the numerical simulation of the forming process.

The stamp forming process of UD FRTPs involves both intra-ply and inter-ply deformation mechanisms [3–9]. Bussetta et al. [6] concluded that intra-ply deformation mechanisms are the most studied mechanisms during the numerical forming of composites. Fig. 1 shows the intra-ply tensile and shear deformation mechanisms during the stamp forming of UD FRTPs. Because UD FRTPs are considered to be sheet materials, deformation along the thickness direction (i.e., transverse

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Nomenclature

T_g, T_m The glass transition temperature, the melting temperature (Celsius temperature)
σ_t, ε_t The tensile stress, tensile strain of the transverse tensile test results
F The load measured from the testing machine
d The change of the length within the gauge length on the transverse tensile test specimen
l_0, A_0 The initial gauge length, initial cross-section area of the transverse tensile or longitudinal shear test specimen
ε_{12}, $\overline{\varepsilon}_{12}$ The in-plane shear strain on the longitudinal shear test specimen, the arithmetic mean value of the in-plane shear strains within the region of interest (ROI)
σ_s, ε_s The shear stress, shear strain of the longitudinal shear test results
$\boldsymbol{\sigma_C}$, $\boldsymbol{\sigma_{PrF}}$, $\boldsymbol{\sigma_{P-F}}$ The stress tensors of the composites, the polymer constituent reinforced by fibres (PrF) part, and the polymer-fibre interface (P-F) part
$\boldsymbol{F_C}$, $\boldsymbol{F_{PrF}}$, $\boldsymbol{F_{P-F}}$ The deformation gradient tensors of the composites, the PrF part, and the P-F part
E_1, E_2, E_3, ν_{12}, ν_{13}, ν_{23}, G_{12}, G_{13}, G_{23} The material constants for the orthotropic elasticity of carbon fibre in ABAQUS
σ, $\overline{\varepsilon}_{pl}$ The flow stress and plastic strain in Johnson-Cook model
A, B, n, m The material constants for the Johnson-Cook model
θ, θ_m, θ_0 The current, melting and the controlling temperature of PA6 (absolute temperature)
ε_{22}^{RVE}, ε_{12}^{RVE} The transverse tensile strain, the longitudinal shear strain of a representative volume element (RVE)
ε_{PrF} The effective strain of PrF
$\varepsilon_{equ-i}^{polymer}$, $\overline{\varepsilon}_{equ}^{polymer}$ The equivalent strain of a polymer element and the averaged equivalent strain of all polymer elements in RVE
dv Local material volume
V_{RVE}, V_p The total volume of the RVE, total volume of polymer elements inside the RVE
V_{F_p}, V_{F_f} The volume fractions of polymer constituent and fibres inside the composite material
k^{map}, $k_{\varepsilon_{22}}^{map}$, $k_{\gamma_{12}}^{map}$ The strain mapping constant, the strain mapping constants for transverse tensile, longitudinal shear deformation
σ_C, σ_{PrF}, σ_{P-F} The stress responses of the composite, the PrF, and the P-F parts
ε_C, ε_{PrF}, ε_{P-F} The strains of composite, the PrF and P-F parts
$\boldsymbol{\sigma_P}$, $\boldsymbol{\sigma_I}$, $\boldsymbol{\sigma_N}$ The stress tensors of the polymer, the intermolecular branch, and the network branch
$\boldsymbol{F_P}$, $\boldsymbol{F_I}$, $\boldsymbol{F_N}$ The deformation gradient tensors of the polymer, the intermolecular and the network branch
$\boldsymbol{F_{Ie}}$, $\boldsymbol{F_{Iv}}$, $\boldsymbol{F_{Ip}}$ The intermolecular elastic, viscous, and plastic deformation gradient tensors
$\boldsymbol{F_{Ne}}$, $\boldsymbol{F_{Nv}}$ The network elastic and viscous deformation gradient tensors
$\boldsymbol{\sigma_{ts}}$, $\boldsymbol{\sigma_{pre-ts}}$ The stress tensors within the tensile and the pre-tensioned springs in the P-F part
$\boldsymbol{F_{ts}}$, $\boldsymbol{F_{pre-ts}}$ The deformation gradient tensors of the tensile and the pre-tensioned springs in the P-F part
ε_1, σ_1, ε_2 The strain, stress of the turning point in the cohesive model, the strain when the cohesive model curve approaches zero
$\sigma_{P-F}(\varepsilon_C)$, $\partial\sigma_{P-F}(\varepsilon_C)/\partial\varepsilon_C$ The cohesive stresses, the first-order derivative of the cohesive model
ε_{10}, $k_{\varepsilon 1}$, σ_{10}, $k_{\sigma 1}$, ε_{20}, $k_{\varepsilon 2}$, $b_{\varepsilon 2}$ The material constants in the cohesive model that related to the temperature dependency
θ_{ref}, θ_g The reference and glass transition temperatures in the P-F part (absolute temperature)
σ_{lim}, σ_{ref}, m_σ The stress limit set on the viscoplastic friction element, the material constants relating to σ_{lim}
ε_{Ie} The equivalent (true) strains of the intermolecular elastic part
ε_{Ne} The equivalent (true) strain of the network elastic part
$\dot{\varepsilon}_{PrF}$ The total equivalent (true) strain rate of the PrF
$\dot{\varepsilon}_{Ie}$, $\dot{\varepsilon}_{Iv}$, $\dot{\varepsilon}_{Ip}$ The equivalent (true) strain rates of the intermolecular elastic, viscous and plastic parts
$\dot{\varepsilon}_{Ne}$, $\dot{\varepsilon}_{Nv}$ The equivalent (true) strain rates of the network elastic and viscous parts
C_1 The temperature-dependent parameter in the intermolecular resistance
E_{ref}, E_1, m_E The temperature-dependent material constants relating to C_1
ν The Poisson's ratio
λ_{Ie} The intermolecular elastic stretch ratio
η_{Iv}, η_{Iv0}, $m_{\theta I}$ The coefficient of viscosity of the dashpot in the intermolecular viscoelastic part, the material constants relating to η_{Iv}
C_p, C_{p1},m_{cp} The temperature-dependent parameter relating to the dashpot in the viscoplastic part, the material constants relating to C_p
$\dot{\gamma}_{01}$, m_{rI}, m_{Iv} The material constants relating to the dashpot in the intermolecular viscoelastic part
$\dot{\varepsilon}_{0p}$, m_{Ip} The material constants relating to the dashpot in the viscoplastic part
C_R, C_{Rref}, C_{r1} The temperature-dependent modulus of the network resistance, the material constants relating to C_R
J_{Ne} The Jacobian determinant associated with the network elastic deformation gradient
$\overline{\lambda}_{Ne}$, $\overline{\lambda}_L$ The average network elastic stretch ratio, the locking stretch
$\mathscr{L}^{-1}$ The inverse of the Langevin function
η_{Nv}, η_{Nv0}, $m_{\theta N}$ The coefficient of viscosity of the dashpot in the network resistance, the material constants relating to η_{Nv}
$\dot{\gamma}_{02}$, m_{rN}, m_{Nv} The material constants relating to the dashpot in the network resistance
Cor The correction coefficient for total equivalent stress considering thermal softening by adiabatic heat
$\dot{\theta}$, θ_{cor} The temperature rate, the corrected temperature during the deformation when the adiabatic heat comes into effect (absolute temperature)
ω The ratio of energy that is converted to temperature change to the total plastic dissipation
m_h The material constant representing the softening parameter
$\overline{C}$, $\overline{C}_{ref}$, $\overline{C}_1$ The temperature-dependent specific heat capacity per unit volume and two related constants

shear) is less significant. Additionally, as fibres are typically considered to be inextensible [10], longitudinal extension is trivial. Therefore, the transverse tension and intra-ply longitudinal shear are identified as the most dominant deformation mechanisms. Transverse tension refers to the tensile deformation that occurs when the prepreg layer is stretched to separate fibres from their original positions [10], while intra-ply longitudinal shear refers to the shear deformation that occurs when fibres slide parallel to each other.

For the characterisation of the intra-ply longitudinal shear properties, many researchers have done extensive research [4,11] and reviews

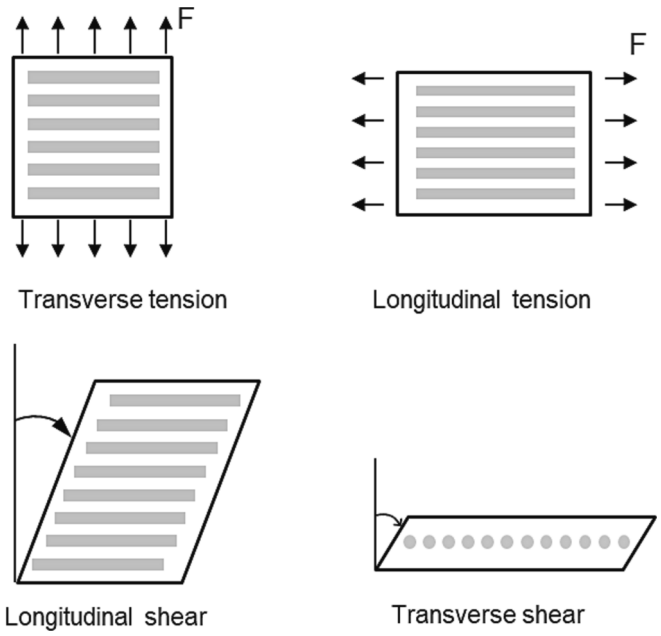


Fig. 1. Intra-ply tensile and shear mechanisms in one prepreg layer of UD FRTPs.

[4,11–13] on the characterisation methods of the shear properties. Fig. 2 summarises the characterisation methods for obtaining the intra-ply shear properties of UD composites employed in previous studies. The most popular methods are the bias-extension test and the picture frame test, both of which were originally invented to characterise textiles [14,15].

Fig. 2a shows the schematic illustrating the deformation mechanism of bias-extension. Regarding its application to UD composites, Potter [13] explored the response of cross-plied (at ± 45°) UD carbon fibre reinforced epoxy (CF/epoxy) via bias-extension test at room temperature. However, the test revealed complex deformation mechanisms, including significant fibre wrinkling and compaction loss. It was concluded that the overall deformation is sensitive to misaligned fibres, which can induce compressive loads and thus fibre buckling. Later, Larberg et al. [16,17] also conducted the bias-extension test on cross-plied CF/epoxy, to characterise the in-plane deformation at different stacking sequences. For the two types of cross-plied CF/Epoxy, although they gained some effective results for HexPly® T700/M21, the behaviour of the other material, Cycom® HTS/977–2, revealed that the inter-ply friction played an important role after the initial in-plane shear dominant deformation. To check the validity of bias-extension tests on molten UD FRTPs, Haanappel et al. [4] conducted bias-extension on UD carbon fibre reinforced poly-ether-ether-ketone (UD CF/PEEK), and the picture in Fig. 2a shows a typical result. Due to the weak integrity of the laminate at the melt state, there was severe material buckling.

Fig. 2b shows the schematic illustrating the deformation mechanism of the picture frame test method. McGuinness et al. [11] conducted this test on cross-plied CF/PEEK laminates with layups of $[0_4/90_4]_{2s}$, under various processing temperatures and strain rates. Although it was concluded that the method was effective in characterising the intra-ply shear properties, there were severe diagonal folds or buckles on the specimens when the enclosed shear angle reached 15-18°, as shown in the picture in Fig. 2b. Haanappel et al. [4] conducted picture frame test on UD CF/PEEK prepregs, and it was observed that ply splitting was easily induced by even small fibre misalignments in the prepregs. Besides, for the picture frame test, an oversized specimen is required to obtain the uniformity in-plane shear deformation. Consequently, a large prepreg size is required, and an equally large environmental chamber is required for the high-temperature characterisation tests. This experimental requirement for a large chamber size is also illustrated in study by Dangora et al. [18] when characterising the intra-ply shear properties of cross-plied FRTPs.

Fig. 2c illustrates the off-axis extension test which is another potential useful method for characterising shear properties. Potter [19]

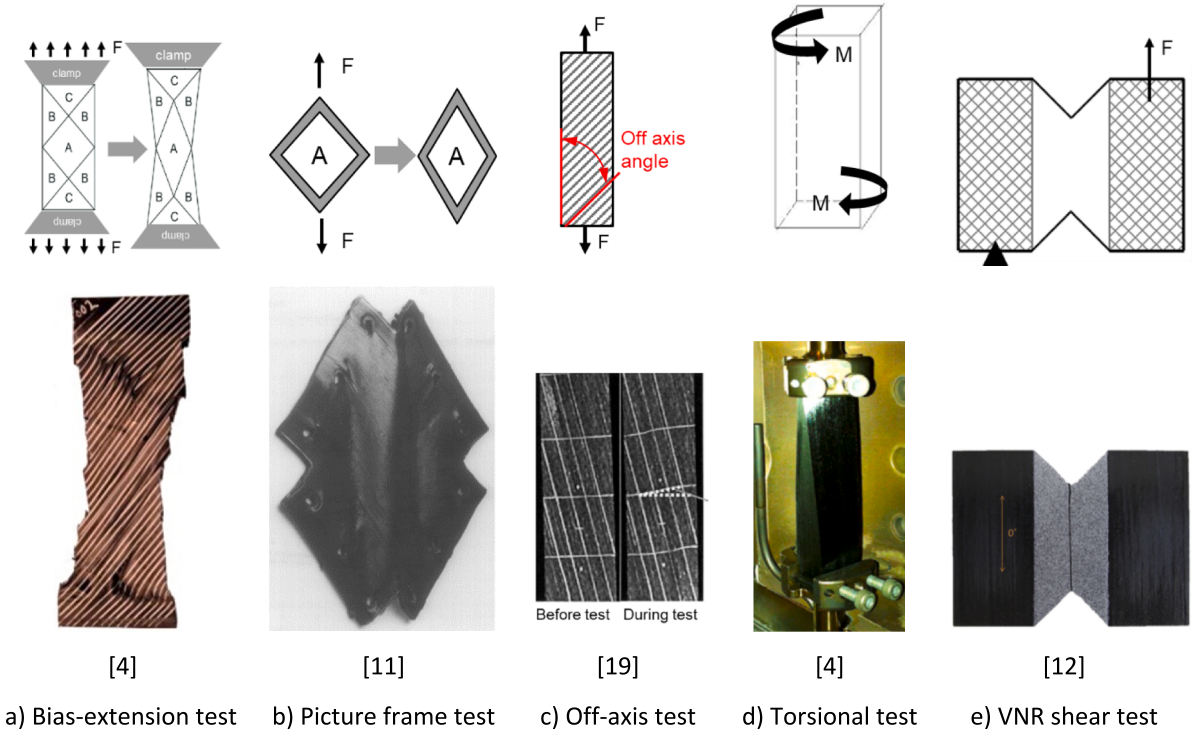


Fig. 2. Characterisation methods for the intra-ply shear properties of UD FRTPs and their corresponding example pictures.

performed off-axis extension tests on UD CF/epoxy at various off-axis angles to determine both transverse and intra-ply shear deformation properties at room temperature and 50 ℃. The specimen before and after deformation is presented in the figure with the white lines drawn along the fibre direction to monitor the fibre rotation. Although the in-plane deformation properties could be obtained through the off-axis extension test, the deformation at various off-axis angles was coupled by both shear and transverse deformations, including transverse tension or compression. Therefore, the coupling effect prevents accurate characterisation of the intra-ply shear properties.

Fig. 2d shows the torsional test proposed by Haanappel et al. [4] for characterising the shear deformation of molten UD FRTPs. The specimen for this torsional test was a prismatic bar with a constant rectangular cross-section, where the fibres were aligned in the direction of the torsional axis. Torque M was applied to one end of the specimen, while the other end was fixed to introduce the longitudinal shear. This torsional experiment setup was also adopted by Dörr et al. [5] on the molten UD carbon fibre reinforced polyamide 6 (UD CF/PA6). In contrast to the experiments by Haanappel et al. [4], a transient load under a single deflection outside the small strain regime was applied [5]. These studies found that both fibre tension and bending were involved in this characterising method, especially in the higher strain regime. Additionally, there was a high requirement for the clamping fixtures to avoid slippage between the fixture and the specimen, which added difficulties to the experiments.

Lastly, Fig. 2e depicts the V-notched rail (VNR) shear test, as defined by ASTM 7078 [20]. As illustrated, there is a vertical load F applied on one side of the specimen and a constraining force on the other side, making the central ligament deform in the manner of simple shear. Tan et al. [12] studied the VNR shear test on UD CF/PEEK at room temperature with various fibre configurations, and an example picture of the specimen at $[0]_{24}$ is shown in the figure. It can be observed from the fracture of the specimen that the in-plane shear deformation was concentrated on the ligament, without having any out-of-plane deformation, as also concluded by the authors [12]. Similarly, Totry et al. [21,22] utilised the VNR shear test to study the in-plane shear properties of UD glass fibre reinforced epoxy (UD GF/epoxy) [22] and UD CF/epoxy [21] at room temperature. Although not widely employed, the VNR shear test method has demonstrated advantages over other methods, such as generating a relatively uniform shear-stress state along the ligament and eliminating edge crushing.

For the characterisation of the transverse tensile properties of UD FRTPs, as discussed earlier, the off-axis extension test characterises the transverse tension for certain off-axis angles [19], but it was difficult to decouple the in-plane transverse tension and shear properties. Wang et al. [23] tested the transverse tensile strength of a UD CF/epoxy specimen at room temperature using a dog-bone specimen on a tensile test machine, but only the maximum force was captured. Margossian et al. [24] designed experiments to directly measure the transverse tensile properties of melt-state UD CF/PA6. They initially explored the option of tensile test on a tensile test machine with an environmental chamber. However, there were difficulties in obtaining results with good reproducibility, mainly due to the thermal expansion of the metallic clamps and their interactions with the specimen. Then, a dynamic mechanical analysis (DMA) system with a constant loading rate on the specimen was used and satisfactory results were obtained for melt-state UD CF/PA6. To the authors' best knowledge, this is the only research that characterised the pure transverse tensile properties of UD FRTPs for the application of thermoforming processes, although it was conducted only at the melt state of the matrix. To simulate the forming process, Dörr et al. [25] utilized the virtual characterization results derived from representative volume element (RVE) studied by Roesner et al. [26] to obtain solid-state properties.

To summarise, there has been a fair number of studies on characterising the intra-ply deformation properties of UD FRTPs. However, most of the characterisation methods have issues in isolating the intra-ply tension and shear deformation mechanisms. Besides, the majority of experiments were performed either at room temperature or above the T_m of the matrices, and no experimental results have been reported at the conditions required for solid-state stamp forming. From a literature analysis, it has been determined that the VNR shear test is suitable for obtaining satisfactory shear properties for UD FRTPs tested under temperatures corresponding to solid-state stamp forming conditions. Additionally, traditional tensile test can be used to characterise the transverse tensile properties, provided that the fixture thermal expansion problem is effectively addressed.

In the aspect of the constitutive models for UD FRTPs, it is crucial to account for the multiscale nature of the material, which involves different physics that occur at different scales, i.e., the macro-scale (the scale of multilayers in a laminate), the *meso*-scale (the scale of one prepreg layer) and the micro-scale (the scale of fibres) [6]. Understanding this multiscale nature is essential for developing accurate constitutive models for UD FRTPs. With this in mind, the choice of modelling scale depends on a balance of model accuracy and computational cost. Normally, macro-scale models for woven FRPs and *meso*-scale models for UD FRPs are used for numerical simulations of stamp forming processes [6]. Both cases require constitutive laws that consider the unique mechanical behaviour of the fibre reinforcements, including the anisotropic behaviour of the warp and weft frames that evolve during transformation due to in-plane shear. Additionally, other behaviours, such as geometric nonlinearities, quasi-inextensibility of the fibres, and strong nonlinear in-plane shear deformation, must be included [9]. To address these specificities, several categories of constitutive models are being studied, including the non-orthogonal elastic model, hypoelastic model, hyperelastic model and the "Ideal Fibre Reinforced Material (IFRM)" model [27]. The non-orthogonal elastic model utilises the elastic stress–strain relationship for the material to examine the impact of fibre strength variations and orientations on the anisotropy during the preforming or fabric draping processes [27–30]. Based on the basic frame of the non-orthogonal elastic model, Peng et al. [31] proposed a non-orthogonal hypoelastic model, which was defined by the relationship between the stress rate and strain rate, for the application of large deformation of woven glass fabrics. In this hypoelastic mode, the global constitutive relation for woven glass fabrics is derived by transforming the contravariant and covariant components to the Cartesian components of the stress and strain tensors. However, the major drawback with the hypoelastic model is that the loading path is unrecoverable, and the hyperelastic models derived from the potential energy function could effectively overcome this drawback [27]. With regards to the formulation of the energy function, Peng et al. [32] decomposed the strain energy function into fibre stretches and yarn cross-over shearing to propose a hyperelastic model that could analyse the large deformation of dry fabric. Later, Gong et al. [33] further refined the work of Peng et al. [32] by proposing an anisotropic hyperelastic model for woven thermoplastic composite prepregs that takes into account the fibres, matrix, and fibre–matrix interaction. They demonstrated its ability to predict the behaviour of woven prepregs during thermoforming processes. Although the above-mentioned constitutive modelling techniques have been used to simulate the stamp forming process of woven FRPs [34], research on the constitutive models of UD FRTPs is still limited. For constitutive models of UD FRTPs, several researchers have employed IFRM for stamp forming applications. The IFRM assumes that the material is incompressible and inextensible in the fibre direction and exhibits Newtonian viscous behaviour for all intra-ply behaviour. This model strategy has proven to be particularly useful since current characterisation studies for stamp forming applications are conducted under the melt state. The IFRM model was firstly introduced by Rogers et al. [35] and later adopted by Haanappel [10] to simulate the stamp forming process of molten UD CF/PEEK. Similarly, Dörr et al. [25] also adopted the IFRM model for molten UD CF/PA6 by assigning rate-dependent hyperelastic models to the intra-ply shear properties, following either a Voigt-Kelvin approach or a

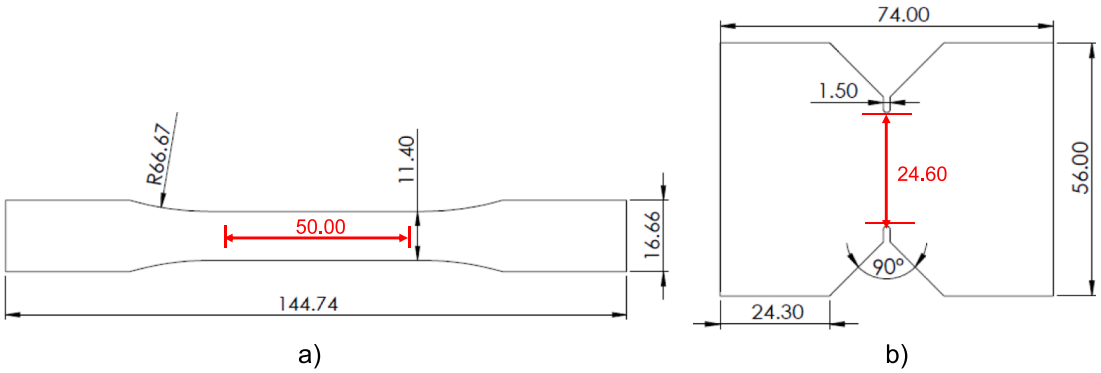


Fig. 3. Experimental specimen designs for a) transverse tensile and b) VNR shear tests. (Unit: mm).

generalised Maxwell approach. The simulation results showed that the Voigt-Kelvin approach was more suitable for predicting the forming behaviour. In addition, the implementation of the IFRM strategy into the commercial software Aniform [10,36] has consequently increased its popularity [37]. However, as indicated by some material characterisation studies [38–40], the thermomechanical properties of UD FRTPs at temperatures close to the T_m of the matrices are more complicated. Therefore, simple elastic, hyperelastic or even viscous-hyperelastic models might not be adequate to describe the complicated temperature-dependent strain hardening or softening material behaviours. To tackle this issue, it is necessary to consider certain physical factors, such as the temperature and strain-rate dependence of the matrices, and the properties of the polymer-fibre interface, to describe the thermomechanical behaviour of UD FRTPs below and close to T_m. Therefore, a constitutive model that is tailored to the unique characteristics of UD FRTPs under solid-state stamp forming conditions is required.

This paper aims to systematically investigate the intra-ply transverse tensile and longitudinal shear properties of UD FRTPs under solid-state stamp-forming conditions through experimental characterisation and constitutive modelling. UD CF/PA6 was selected for the study, as PA6 is a representative automotive grade polymer matrix experiencing growing importance in lightweight automotive applications. Tests were conducted at temperatures ranging from the glass transition temperature T_g to the melting temperature T_m of the matrix material, and the strain rate was controlled between 0.001 and 0.4 /s, in accordance with the solid-state stamp forming conditions. The transverse tensile and longitudinal shear properties were characterised via tensile test and VNR shear test, respectively. A new physically based constitutive model, partially based on the authors' previous study on constitutive modelling of thermoplastics (TPs) [40], was proposed to bridge the TP matrices and UD FRTPs. The new model was utilised to predict the in-plane deformation of the UD FRTPs and provided further insights into their deformation mechanisms. It is important to note that the scale of this research focuses on the *meso*-scale level [6], specifically, at the single prepreg level.

2. Experimental design

2.1. Materials and specimen design

The material studied in this paper is Cetex® TC910 UD carbon fibre reinforced PA6 tapes purchased from Toray Advanced Composites [41]. The PA6 resin content by weight is 40 %. The UD tape prepreg layers with a configuration of $[0]_{24}$ were consolidated inside a pair of metal moulds using a hot press. The consolidation temperature was 250 ± 10

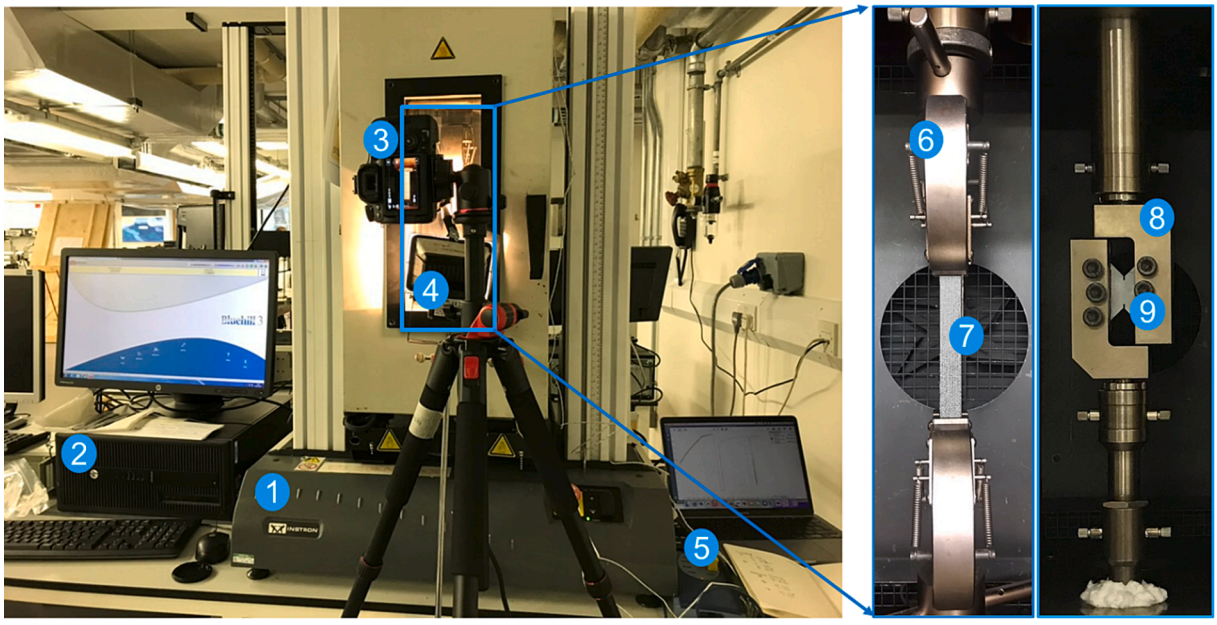


Fig. 4. Experimental setup. ① Instron 3369 tensile testing machine with an environmental chamber; ② PC and BLUEHILL software; ③ Camera; ④ Lamp; ⑤ Thermocouples and datalogger; ⑥ Tensile test rig; ⑦Transverse tensile test specimen; ⑧ VNR shear test rig; ⑨ VNR shear test specimen.

Table 1
Test matrices for transverse tensile and VNR shear tests.

Temperature (℃)	Strain rate (/s)						
	Transverse tension				VNR shear		
	0.001	0.01	0.1	0.25	0.004	0.04	0.4
180		√				√	
200	√	√	√	√	√	√	√
210		√				√	
220		√				√	

℃. The pressure was set to 2 bar at the heat-up stage; and when the target temperature was reached, the consolidation pressure was set to 4 bar for 20 min. Subsequently, the laminate was cooled to 80 ℃ under 4 bar. The initial thickness of one prepreg layer was 0.16 mm. After consolidation, the formed laminates were at a thickness of 3.3 mm. The consolidation quality of these laminates was examined using a portable 'Prisma TOFD' ultrasonic C-scan device supplied by Sonatest Ltd. UK [42,43]. All laminates produced were free from defects and were cut into the transverse tensile and VNR shear test specimens via water jet cutting.

The transverse tensile specimen was designed according to the aspect ratio of the dog bone specimen described in ASTM 638 [44,45]. With the consideration of the total length of the stretched specimen that the environmental chamber can accommodate, the dimensions are proportionally scaled down accordingly. The detailed dimensions of the transverse tensile specimen are presented in Fig. 3a, and the gauge length was defined as 50.00 mm. As for the VNR shear specimen, the initial design was according to the ASTM 7078 standard [20]. In the preliminary tests, however, it was difficult to concentrate stress along the ligament, resulting in poor repeatability. Therefore, a modification was made by adding two notches with a width of 1.5 mm (which was determined by the width of the water jet cutter) and a depth of 3 mm each, and the final specimen design is shown in Fig. 3b. The ligament length of the VNR shear specimen after modification was 24.60 mm. All the experimental specimens were pre-dried at 70 ℃ for 7 h to mitigate deconsolidation during heating in the test [46]. After the pre-drying, the specimens were tested within 24 h.

2.2. Experimental setup and procedure

Fig. 4 shows the experimental setup on an Instron 3369 tensile testing machine controlled by the BLUEHILL software. An environmental chamber was mounted on the machine to create the desired testing temperatures. During experiments, thermocouples were used to monitor the temperatures on the surface of the specimen and the clamping rig, and the temperatures were recorded by a datalogger. A camera was set up for recording the sample deformation during the test at a rate of 60 frame-per-second (fps). The recorded video was later used to extract the strain. A lamp was set to provide sufficient light for optimal image quality. The testing rig for the transverse tensile test was the Instron standard tensile clamp; and for the VNR shear test, it was a pair of bespoke shear clamps made of titanium alloy. For each test, a procedure during heating was strictly followed to mitigate the potential compression that comes from the thermal expansion of the specimen and tools. First, the environmental chamber together with the metal clamps was heated to the desired temperature without the specimen. Then, the specimen was mounted by clamping on only one side. After that, the specimen was heated to the desired temperature and then soaked for 10 min to achieve a uniform temperature. Later, the door of the environmental chamber was opened, and the other side of the specimen was quickly tightened. Following that, the door of the environmental chamber was closed again to heat the entire system. After the specimen was heated to the desired temperature again and soaked for another 20 min, the experiment started. Through this process, the compression load was monitored to be no more than 20 N.

The test conditions employed in this study were determined according to solid-state stamp-forming conditions. For each test condition, three tests were conducted to check the repeatability. The detailed test matrices for transverse tensile and VNR shear tests are listed in Table 1.

2.3. Strain measurement and data postprocessing

Both the tensile and shear strains were measured using the digital image correlation (DIC) technique. Before the experiments, a random speckle pattern was created on each specimen with white paint for the background and black paint for the speckles. The deformation images were processed by a 2-dimensional (2D) commercial software, GOM, Germany. For each recorded deformation, a facet size of 16 x 16 pixels with a distance of 14 pixels was used in the software.

For the transverse tensile test specimen, the tensile stress, σ_t, and tensile strain, ε_t, were defined as Eqs. (1)-(2), respectively.

$$\sigma_t = \frac{F}{A_0}\cdot(1 + d/l_0) \tag{1}$$

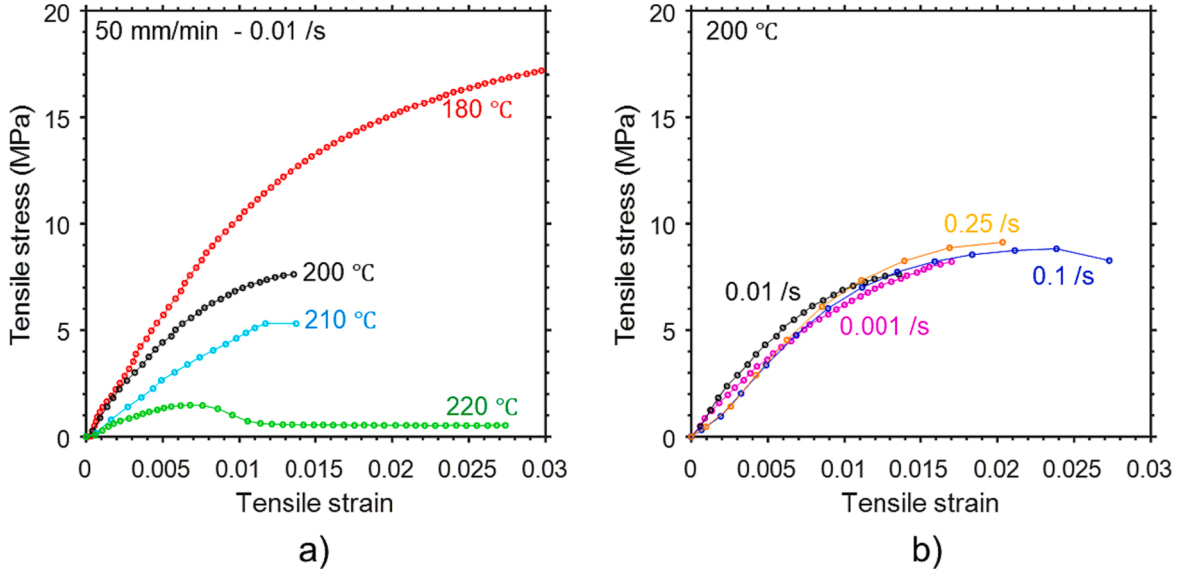


Fig. 5. Experimental results of transverse tensile tests on UD CF/PA6 under a) a temperature range of 180 – 220 ℃, at a strain rate of 0.01 /s and b) a strain rate range of 0.001 – 0.25 /s, at a temperature of 200 ℃.

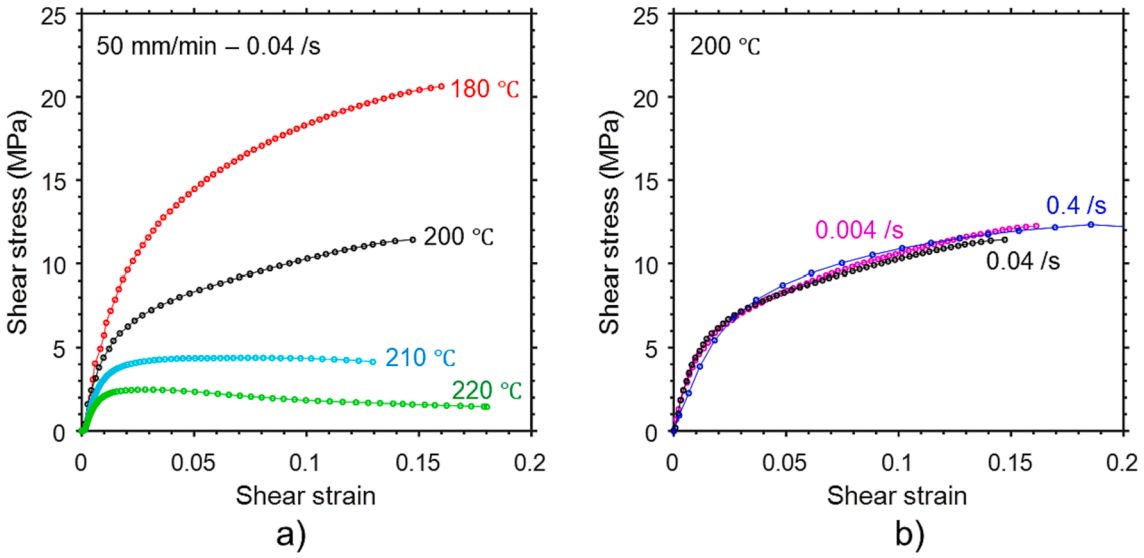


Fig. 6. Experimental results of VNR shear tests on UD CF/PA6 under a) a temperature range of 180 – 220 ℃, at a strain rate of 0.04 /s and b) a strain rate range of 0.004 – 0.4 /s, at a temperature of 200 ℃.

$$\varepsilon_t = \ln(1 + d/l_0) \tag{2}$$

where F is the load measured from the testing machine, d is the change of the length within the gauge length area measured by the GOM software, $l_0 = 50.00$ mm is the initial gauge length, A_0 is the initial cross-section area of the transverse tensile test specimen.

For the VNR shear test, the shear stress, σ_s, and shear strain, ε_s, were defined as Eqs. (3)-(4), respectively.

$$\sigma_s = \frac{F}{A_0} \tag{3}$$

$$\varepsilon_s = \bar{\varepsilon}_{12}; \tag{4}$$

where F is also the load measured by the load cell of Instron. And A_0 is the initial cross-section area of the VNR shear test specimen. The shear strain field was analysed via DIC by the GOM software. A section was constructed in the GOM software along the ligament of the specimen, acting as the region of interest (ROI) [47]. The term $\bar{\varepsilon}_{12}$ is the arithmetic mean value of the in-plane shear strains ε_{12} of all facets across the ROI.

3. Experimental results and analysis

3.1. Experimental results of the transverse tensile and VNR shear tests

The tensile stress–strain curves of transverse tensile tests are shown in Fig. 5. It is evident that a linear elastic region exists at the beginning of the deformation. After the initial linear elastic deformation, nonlinear deformation regions appear, where strain-hardening or -softening phenomena occur. It is crucial to note that in this research, the term "strain-hardening" refers to the increase in stress with increasing strain, whereas the term "strain-softening" denotes a decrease in stress with increasing strain. Fig. 5a demonstrates that temperature has a significant impact on the stress–strain behaviour (at a tensile strain rate of 0.01 /s). Specifically, an increase in temperature results in a notable decrease in stress at the same tensile strain level. Within the tested temperature region (180 – 220 ℃), at lower temperatures (180/200 ℃), there is strain-hardening behaviour at the nonlinear deformation region. At higher temperatures (210/220 ℃), however, strain-softening phenomena are observed. At 220 ℃, the stress level gradually reaches a plateau

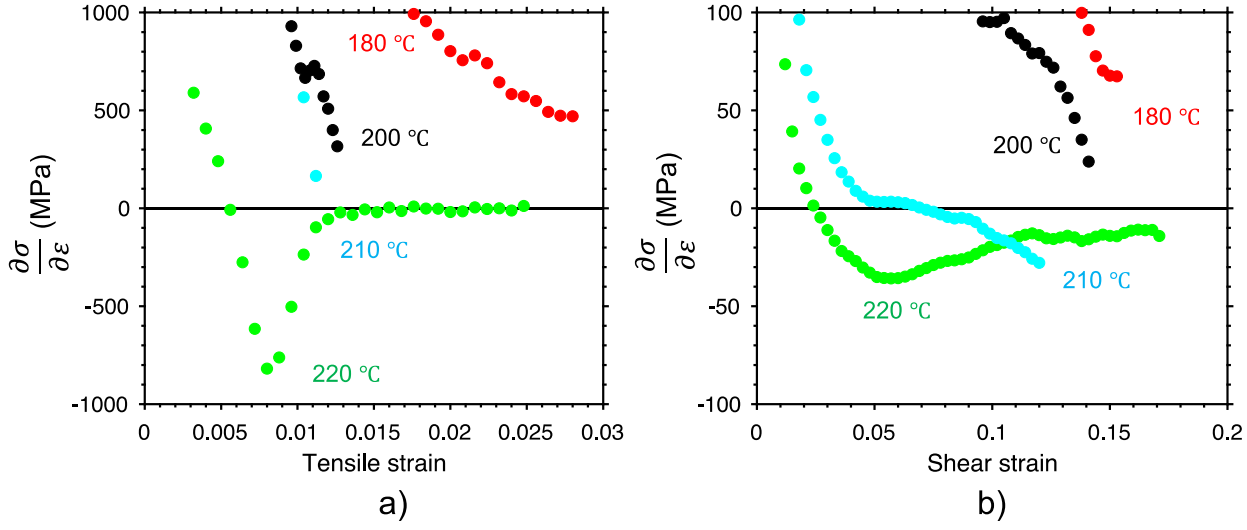


Fig. 7. The tangent modulus versus strain at different temperatures (180 – 220 ℃) under a) transverse tensile at a strain rate of 0.01 /s, only the data within a range of ± 1000 are shown, and b) VNR shear tests at a strain rate of 0.04 /s, only the data within a range of ± 100 are shown.

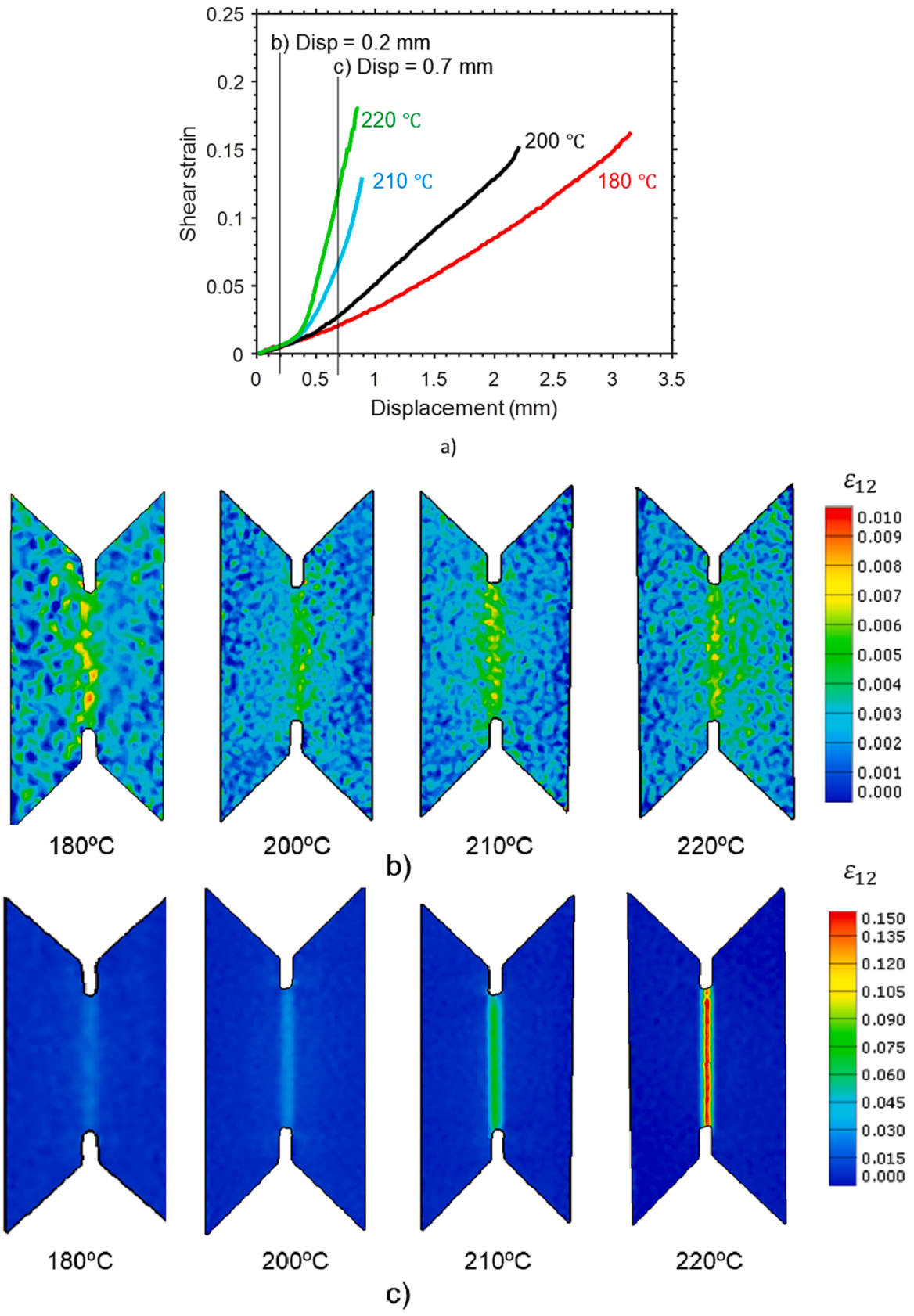


Fig. 8. Localised deformation analysis of VNR shear tests under the temperature range of 180 – 220 ℃. a) Shear strain evolution versus the crosshead displacement, and shear strain distribution at the displacement of b) 0.2 mm and c) 0.7 mm at different temperatures.

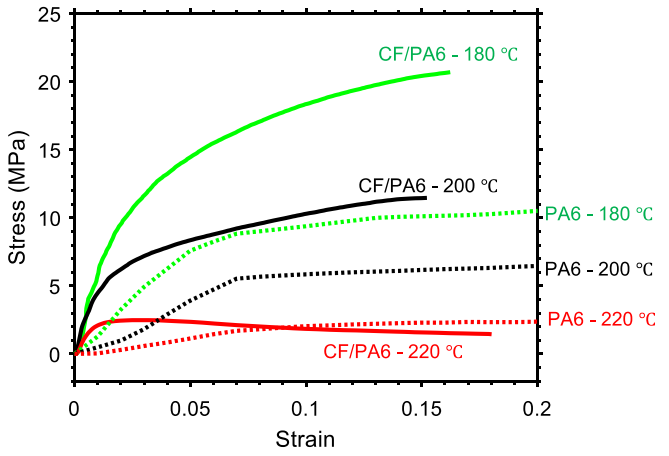


Fig. 9. Stress–strain curves of UD CF/PA6 under longitudinal shear compared to PA6 under uniaxial tension [40], under various temperatures (180 – 220 ℃).

after the period of strain softening. On the other hand, Fig. 5b indicates that the strain-rate effects are marginal within the tested conditions (200 ℃, 0.001 – 0.25 /s). Moreover, it is evident that the specimen fractures easily during the transverse tensile tests, as indicated by the relatively small strains at failure in Fig. 5.

Fig. 6 shows the stress–strain curves of VNR shear tests under various temperatures and strain rates. Fig. 6a illustrates the temperature effects on the longitudinal shear behaviour. Similar to transverse tensile deformation, the temperature effects (at a strain rate of 0.04 /s) significantly impact longitudinal shear behaviour, leading to a decrease in stress level with increasing temperature. An elastic deformation region exists at the beginning of deformation, followed by strain-hardening or -softening behaviour. At lower temperatures (180/200 ℃), stress increases consistently with the increasing shear strain, indicating continuous strain hardening. At higher temperatures (210/220 ℃), obvious strain-softening behaviour occurs. Fig. 6b shows that the strain-rate effects at 200 ℃ with strain rate ranging from 0.004 to 0.4 /s are insignificant on the longitudinal shear properties.

There are noticeable strain-softening effects observed at the starting temperature of 210 ℃ for both transverse tensile and VNR shear tests. Additionally, it is noteworthy that the material exhibits a decreasing trend in strain-to-failure with increasing temperature at temperatures below or equal to 210 ℃. However, this trend disappears when the temperature approaches T_m, i.e., 220 ℃. To gain a further understanding of the strain-hardening and -softening behaviour, the transverse tensile and longitudinal shear properties are further analysed in the following sections.

3.2. Analysis of the intra-ply deformation properties

The strain hardening or softening behaviour can be inferred clearly by the tangent modulus evolution as indicated in Fig. 7. The tangent modulus is defined as the derivative of the stress–strain curve. Therefore, according to the definition, after the initial elastic deformation, the positive tangent moduli indicate strain hardening, while the negative tangent moduli indicate strain softening. It is found that for both transverse tensile and VNR shear tests, at lower temperatures (180/200 ℃), there is only strain hardening, although the tangent modulus representing the level of strain hardening gradually decreases during deformation; whereas at higher temperatures (210/220 ℃), strain softening can be observed after certain deformation. Besides, strain softening occurs earlier at the higher temperature (220 ℃), which indicates easier localised deformation.

To gain deeper insight into the material's strain-hardening/-softening responses and corresponding deformation mechanisms, further analysis was conducted and illustrated through Fig. 8. VNR shear

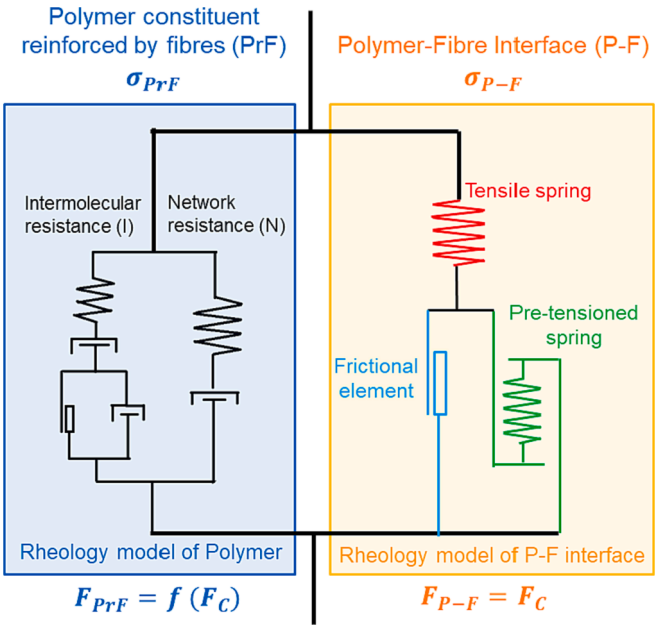


Fig. 10. The rheology scheme of the constitutive model.

results were selected for the analysis, because the material could achieve a much greater level of deformation under shear that is deemed the dominant deformation mode during forming. The nonlinear relationship between the shear-strain evolution and the applied displacement on the specimen, as shown in Fig. 8a, can be used to infer localised deformation. The shear strains are similar for different test conditions until the displacement reaches approximately 0.4 mm, after which the strains in the ligament area increase more rapidly with increasing temperature, indicating greater levels of localised deformation.

Fig. 8b and c depict the shear strain fields across the specimen at displacements of 0.2 and 0.7 mm, respectively, for varying testing temperatures (180 – 220 ℃). In Fig. 8b, at a displacement of 0.2 mm, the shear strain fields show a similar distribution across different temperatures. However, in Fig. 9 shows a comparison between the stress–strain properties of the UD CF/PA6 and PA6 polymer at different temperatures. The stress–strain curves of UD CF/PA6 are the longitudinal shear stress–strain curves, and the stress–strain curves of PA6 under uniaxial tension come from the authors' previous study [40]. Note that although UD CF/PA6 was tested at 0.01 /s and PA6 was tested at 0.1 /s, the comparison can still provide valid indications because it has shown that the strain-rate effects on both materials are insignificant within the temperature and strain-rate ranges considered in this work. As can be seen, the stress level of UD CF/PA6 is higher than PA6, at 180 and 200 ℃, and for the early stage of deformation at 220 ℃. These results reflect the stiffening effect attributed to the fibre reinforcement. Additionally, in Fig. 9, no significant softening behaviour is observed in PA6 under all tested temperatures (180 – 220 ℃), while there is a noticeable strain softening observed after the initial deformation stage in UD CF/PA6 at 220 ℃. When discussing the possible source of this strain-softening behaviour, all constituents in the material should be taken into consideration. Therefore, in the case of UD CF/PA6, the constituents that contribute to the total strength are PA6, reinforcing carbon fibres (CFs), and the interface between CFs and PA6. Because CFs can be considered rigid bodies and PA6 shows no strain-softening behaviour, it is reasonable to conclude that the strain-softening behaviour comes from the degradation of the fibre–matrix interface. These findings informed the development of the physically based constitutive model of UD FRTPs in Section 4.

Table 2
Material constants for carbon fibre and PA6 matrix material input in RVE.

Carbon fibre Orthotropic elasticity		Matrix (PA6) Linear elasticity and Johnson-Cook plasticity	
E_1(MPa)	214,000	E(MPa)	152.77 (180 ℃); 70.02 (200 ℃); 52.91 (210 ℃); 27.79 (220 ℃).
E_2(MPa)	26,000	ν	0.48
E_3(MPa)	26,000	A(MPa)	11.85
ν_{12}	0.28	B(MPa)	14.53
ν_{13}	0.28	n	0.73
ν_{23}	0.45	m	1.08
G_{12}(MPa)	112,000	θ_m(K)	503
G_{13}(MPa)	112,000	θ_0(K)	433
G_{23}(MPa)	8,996.5		

The equation form of Johnson-Cook model is:

$\sigma = \left[A + B(\bar{\varepsilon}_{pl})^n\right] \cdot \left(1 - \left(\frac{\theta - \theta_0}{\theta_m - \theta_0}\right)^m\right)$, where σ and $\bar{\varepsilon}_{pl}$ are the flow stress and plastic strain, respectively. The terms A, B, n, m are material constants for the Johnson-Cook model. Terms θ, θ_m and θ_0 are the current temperature, melting temperature and the controlling temperature, respectively, all at unit of Kelvin.

4. Constitutive modelling

In this section, a novel physically based constitutive model is presented. The new model comprises basic rheology elements to represent the viscous-elastic–plastic deformation properties of the composite material. The rheology of the model is shown in Fig. 10, where the model for calculating the stress from deformation is composed of two parts: one part represents stress contributed by the polymer constituent reinforced by fibres (PrF), and the other part presents stress contributed by the polymer-fibre interface (P-F). Note that the model does not consider the strain-rate effects because they were found to be negligible in the interested range.

The overall constitutive model is indicated as Eqs. (5)-(6)

$$\boldsymbol{\sigma}_C(\boldsymbol{F}_C) = \boldsymbol{\sigma}_{PrF}(\boldsymbol{F}_{PrF}) + \boldsymbol{\sigma}_{P-F}(\boldsymbol{F}_{P-F}) \tag{5}$$

$$\boldsymbol{F}_{PrF} = \boldsymbol{f}(\boldsymbol{F}_C); \boldsymbol{F}_{P-F} = \boldsymbol{F}_C \tag{6}$$

where the form $\boldsymbol{\sigma}(\boldsymbol{F})$ denotes the stress response with respect to a given deformation of $\boldsymbol{F}$. $\boldsymbol{\sigma}_C$, $\boldsymbol{\sigma}_{PrF}$ and $\boldsymbol{\sigma}_{P-F}$ are the stress tensors of the composite, the PrF part, and the polymer- P-F part, respectively. $\boldsymbol{F}_C$, $\boldsymbol{F}_{PrF}$ and $\boldsymbol{F}_{P-F}$ are the deformation gradient tensors corresponding to the composites, the PrF part, and the P-F part, respectively. The form $\boldsymbol{f}(\boldsymbol{F}_C)$ represents that $\boldsymbol{F}_{PrF}$ is a function of $\boldsymbol{F}_C$, based on a hypothesis that there are mapping relationships between the overall deformations of the composite and the effective deformations of the polymer constituent. The investigation of the mapping relationships is detailed in the following subsection.

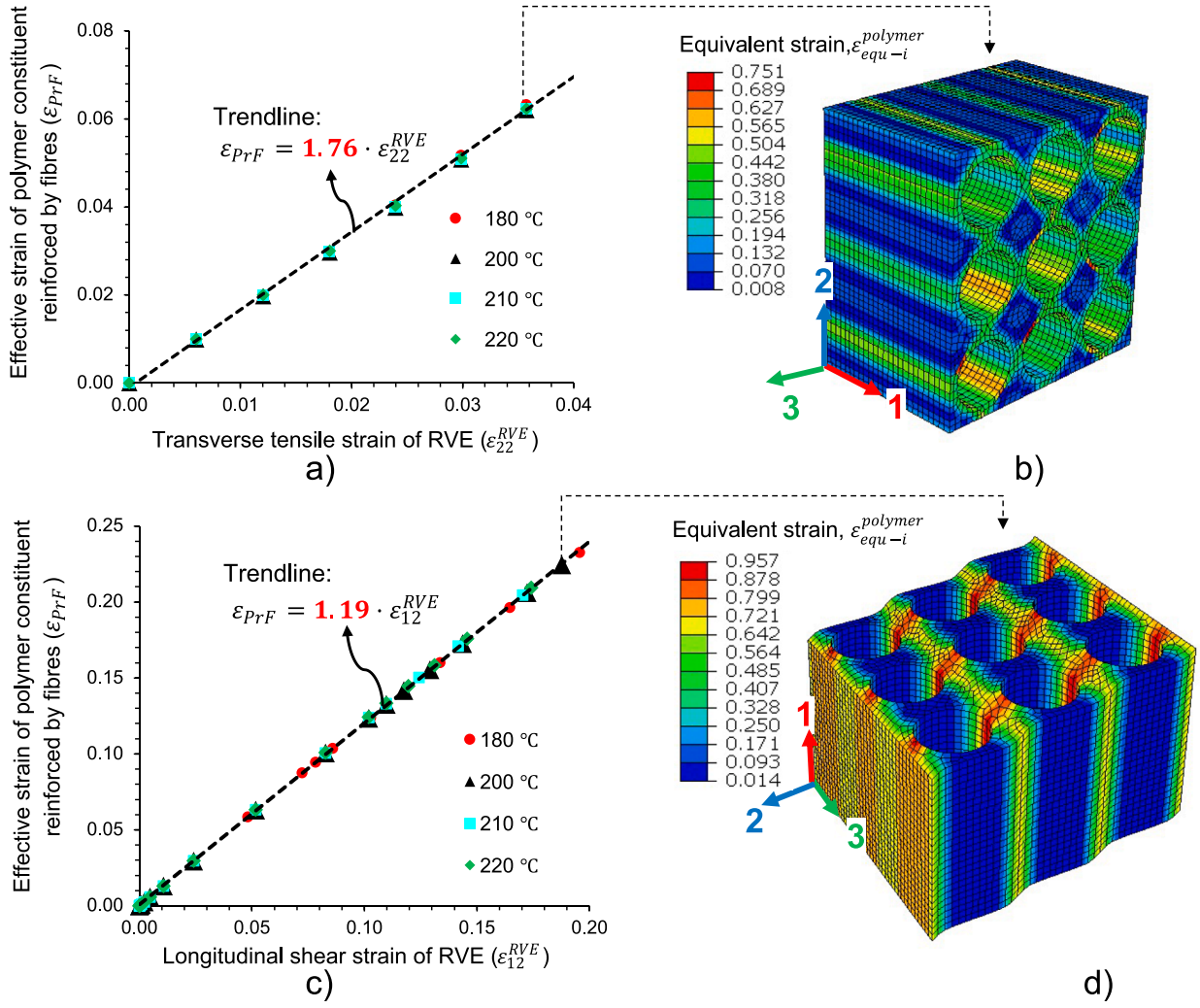


Fig. 11. Analysis of the mapping relationships between the composite deformations and the effective deformations of PrF using RVE. a) ε_{PrF} vs. ε_{22}^{RVE} under transverse tensile at various temperatures (180——220 ℃), and b) the equivalent strain distribution in polymer elements when $\varepsilon_{22}^{RVE} = 0.036$, at 200 ℃. c) ε_{PrF} vs. ε_{12}^{RVE} under longitudinal shear at various temperatures (180——220 ℃) and d) the equivalent strain distribution in polymer elements when $\varepsilon_{12}^{RVE} = 0.188$, at 200 ℃.

4.1. The contribution of the polymer constituent reinforced by fibres (PrF)

One crucial question that needs to be addressed when determining the contribution of PrF is the extent of deformation that occurs in the polymer constituent compared to the composite deformation. Given the difference in stiffness between the polymer and composite materials, the polymer constituent is likely to experience more deformation than the composite as a whole. As a result, it is necessary to establish a mapping relationship that describes the relative relationship between the deformation of the polymer constituent compared to the total deformation of the composite. To calculate the specific deformation of the polymers within the composite, the representative volume element (RVE) technique is used, which involves creating an FEA model that comprises the basic microstructure of the composite. In a composite laminate, the fibres are randomly distributed. For simplicity, however, the periodic square array arrangement of fibres is used [48]. In this case, the researchers have constructed a RVE model with nine unidirectional fibres that are uniformly distributed within the polymer matrix. The RVE model is then simulated using ABAQUS, a FEA software.

In this RVE model, the fibre volume fraction (V_{Ff}) was 46.87 %, corresponding to the datasheet of the UD CF/PA6 prepreg [41]. The fibre material was set to be orthotropic elastic, using the parameters for AS4 carbon fibres obtained from a reference [49]. For the matrix material (PA6), linear elasticity followed by the Johnson-Cook plasticity constitutive model were used [47], and the material constants were determined from the experimental results previously obtained by the authors [40]. The summary of material constants used for the RVE study are listed in Table 2.

Two periodic boundary conditions (PBCs) were added on the RVE model, corresponding to the transverse tension and longitudinal shear deformation, respectively. The PBCs were implemented using an ABAQUS plugin called EasyPBC, developed by Omairey et al. [50]. The transverse tensile strain, ε_{22}^{RVE}, and longitudinal shear strain, ε_{12}^{RVE}, were obtained also according to the definition provided in reference [50].

From simulations on this RVE model, the strain in polymer elements, with respect to the overall composite deformation, can be obtained. Due to fibre reinforcement, the strain varies in polymer elements under both transverse tensile and longitudinal shear conditions, as shown in Fig. 11b and d. An effective strain value representing the deformation of PrF, ε_{PrF}, is needed. The value of ε_{PrF} is determined by integrating the local strain corresponding to the material volumes, dv, and dividing the results by the total volume of the RVE, V_{RVE}. The mathematical expression of ε_{PrF} can be denoted by Eq. (7).

$$\varepsilon_{PrF} = \frac{\int \varepsilon_{equ-i}^{polymer} dv}{V_{RVE}} \tag{7}$$

where $\varepsilon_{equ-i}^{polymer}$ represents the equivalent strain of a local polymer element in RVE.

By introducing $\overline{\varepsilon}_{equ}^{polymer}$, representing the averaged equivalent strain of all polymer elements in the RVE, Eq. (7) can be simplified as follows:

$$\varepsilon_{PrF} = \frac{\overline{\varepsilon}_{equ}^{polymer} \cdot V_p}{V_{RVE}} = \overline{\varepsilon}_{equ}^{polymer} \cdot V_{F_p} \tag{8}$$

where V_p denotes the total volume of polymer elements inside the RVE, and V_{F_p} denotes the volume fraction of polymer constituent in the RVE.

The simulation results and mapping analysis of the RVE are presented in Fig. 11. For the clarity of the mapping analysis:

- ε_{22}^{RVE} represents the total strain of composite under transverse tension.
- ε_{12}^{RVE} represents the total strain of composite under longitudinal shear.
- ε_{PrF} represents the equivalent strain of the polymer constituent.

Fig. 11a and c depict the relationship between ε_{PrF} versus (vs.) ε_{22}^{RVE} and ε_{12}^{RVE}, respectively. Fig. 11b and d display examples of equivalent strain ($\varepsilon_{equ-i}^{polymer}$) fields for the polymer elements under transverse tension and longitudinal shear, respectively, both at 200 ℃. The specific deformation stages are indicated in the figures, with values $\varepsilon_{22}^{RVE} = 0.036$ and $\varepsilon_{12}^{RVE} = 0.188$. To illustrate the deformation of polymers, the fibres are excluded from the view. Notably, the equivalent strains of the polymer elements, $\varepsilon_{equ-i}^{polymer}$, show a certain range under both deformation modes, with values ranging from 0.008 to 0.751 for transverse tension and 0.014 to 0.957 for longitudinal shear. From Fig. 11a and c, it is evident that ε_{PrF} exhibits linear relationships with both ε_{22}^{RVE} and ε_{12}^{RVE}, independent of temperature across the studied range (180 to 220 ℃). This linear relationship is demonstrated by the presence of trendlines in Fig. 11a and c. As a result, the constant ratios of ε_{PrF} to ε_{22}^{RVE} or ε_{12}^{RVE}, referred to as strain mapping factors, k^{map}, can be extracted as the slopes of their trendlines. It is clearly indicated that for transverse tension and longitudinal shear deformation, $k_{\varepsilon_{22}}^{map} = 1.76$ and $k_{\varepsilon_{12}}^{map} = 1.19$, respectively.

With the polymer mapping relationship obtained, the stress contribution of fibre reinforced polymers can be calculated according to the polymer deformation. Therefore, Eqs. (5) and (6) can be revised as Eq. (9) in a one-dimensional form:

$$\sigma_C(\varepsilon_C) = \sigma_{PrF}(\varepsilon_{PrF}) + \sigma_{P-F}(\varepsilon_{P-T}) = \sigma_{PrF}(k^{map} \cdot \varepsilon_C) + \sigma_{P-F}(\varepsilon_C) \tag{9}$$

where the form $\sigma(\varepsilon)$ denotes the stress response with respect to a given strain of ε. σ_C, σ_{PrF} and σ_{P-F} represent the stress responses of the composite, the PrF, and the P-F parts, respectively; ε_C, ε_{PrF} and ε_{P-F} are composite strain and strains corresponding to the PrF and P-F parts, respectively. k^{map} represents $k_{\varepsilon_{22}}^{map}$ or $k_{\varepsilon_{12}}^{map}$, depending on whether the deformation is transverse tension or longitudinal shear. The key finding from the analysis validates the hypothesis on the mapping relationships between the composite deformations and the effective deformations of the PrF. This mapping relationship means that when the composite strain is ε_C, the contribution of the PrF equals to the stress generated in the polymer when the polymer strain is $k^{map} \cdot \varepsilon_C$.

The polymer constitutive model recently developed by the authors [40] can be adopted to calculate σ_{PrF}. The constitutive model is briefly introduced as below.

The polymer model comprises two branches: an intermolecular branch (I) that processes viscoelastic-viscoplastic behaviour to represent the intermolecular stretching within polymers, and a network branch (N) that processes viscoelastic behaviour to represent the network stretching. The constitutive model can be represented as Eqs. (10)-(11)

$$\boldsymbol{\sigma_P} = \boldsymbol{\sigma_I} + \boldsymbol{\sigma_N} \tag{10}$$

$$\boldsymbol{F_P} = \boldsymbol{F_I} = \boldsymbol{F_N}; \boldsymbol{F_I} = \boldsymbol{F_{Ie}} \cdot \boldsymbol{F_{Iv}} \cdot \boldsymbol{F_{Ip}}; \boldsymbol{F_N} = \boldsymbol{F_{Ne}} \cdot \boldsymbol{F_{Nv}} \tag{11}$$

where $\boldsymbol{\sigma_P}$, $\boldsymbol{\sigma_I}$ and $\boldsymbol{\sigma_N}$ are the stress tensors of the polymer, the intermolecular branch, and the network branch, respectively. $\boldsymbol{F_P}$, $\boldsymbol{F_I}$ and $\boldsymbol{F_N}$ are the deformation gradient tensors of the polymer, the intermolecular branch, and the network branch, respectively. $\boldsymbol{F_{Ie}}, \boldsymbol{F_{Iv}}, \boldsymbol{F_{Ip}}$ are the elastic, viscous and plastic deformation gradient tensors in the intermolecular branch, respectively, and $\boldsymbol{F_{Ne}}$ and $\boldsymbol{F_{Nv}}$ are the elastic and viscous deformation gradient tensors in the network branch, respectively. In the intermolecular and network branches, the stress tensors ($\boldsymbol{\sigma_I}$ and $\boldsymbol{\sigma_N}$) are calculated from a Neo-Hookean hyperelastic spring and an Eight-chains hyperelastic spring, respectively. In this study, $\boldsymbol{\sigma_P}$ is equivalent to $\boldsymbol{\sigma_{PrF}}$, and $\boldsymbol{F_P}$ is equivalent to $\boldsymbol{F_{PrF}}$ which can be calculated from the composite deformation $\boldsymbol{F_C}$ and the mapping relationship k^{map}.

4.2. The contribution of the polymer-fibre interface (P-F)

The contribution of the P-F to the constitutive model is based on the

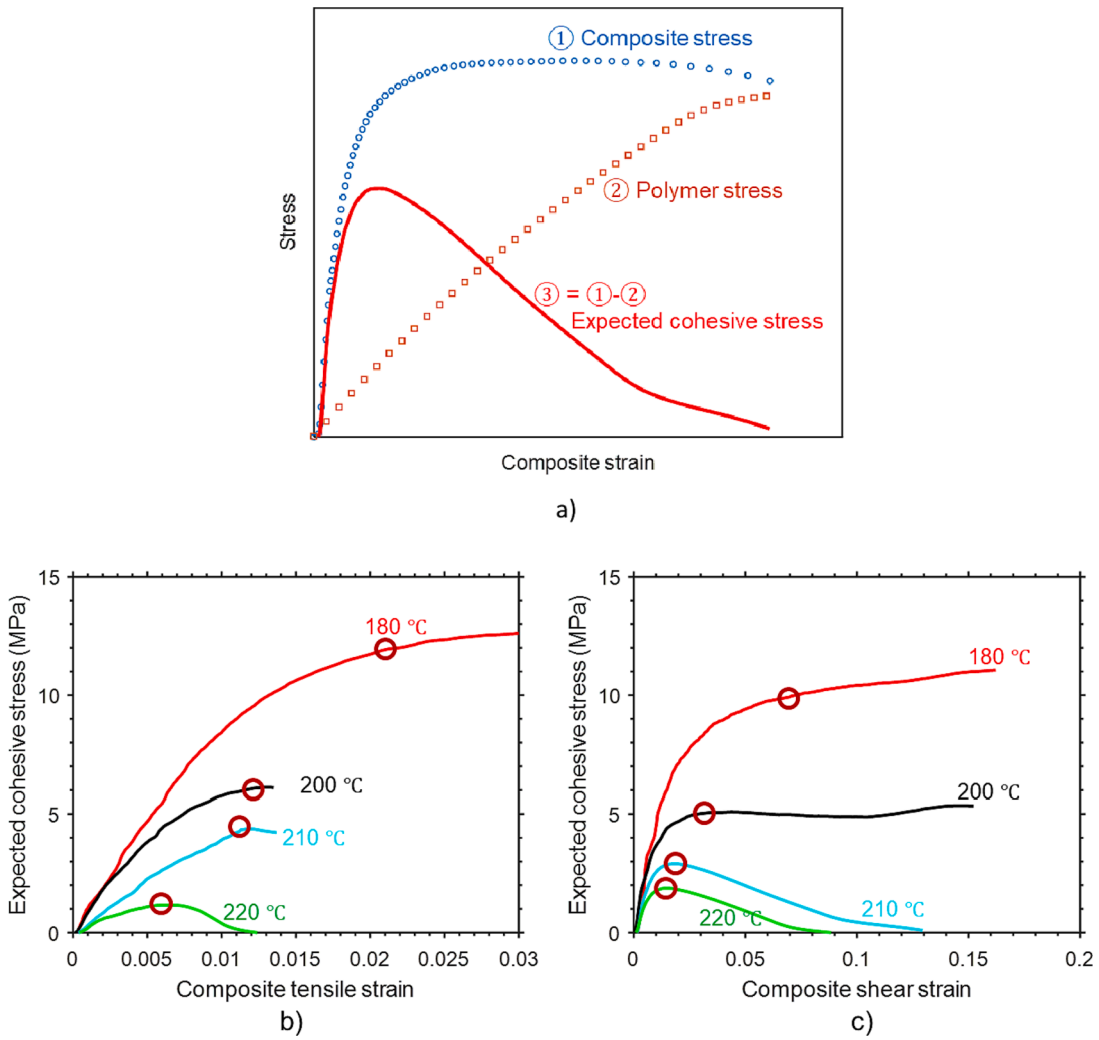


Fig. 12. a) the calculation method for obtaining the expected cohesive stress curve. the expected cohesive stress at different temperatures (180 – 220 ℃) under b) transverse tension and c) longitudinal shear deformation (circles representing the turning point of each curve).

adoption of the cohesive stress, that is commonly used to describe interface properties. Cohesive stress refers to the force required to separate individual atoms for a prescribed interatomic separation distance [51]. For this study, expected cohesive stress is defined to represent the P-F contribution. Fig. 12a shows the process of obtaining the expected cohesive stress (curve ③): by subtracting the polymer stress after mapping adjustment (curve ②) from the composite stress (curve ①) according to Eq. (9). The horizontal axis in Fig. 12a is the composite (tensile/shear) strain instead of the separation distance typically used in cohesive theories [51] for the simplicity of later analysis. Theoretically, there will also be mapping relationships between the separation distance and composite deformation, but that would not be the focus of this study. Fig. 12a indicates that the composite stress in the initial phase is primarily attributed to P-F. This is largely a result of stress concentration at the polymer-fibre interfaces during deformation. As Huang et al. [52] highlighted, stress concentration is more prevalent at the polymer-fibre interface than within the fibres or matrix. Furthermore, various interfacial bonding mechanisms exist, such as molecular entanglement, element inter-diffusion, electrostatic attraction, and chemical reactions between reinforcement and matrix surfaces, etc. As noted by Kim et al. [53], these mechanisms result in a robust physical force, making the separation of reinforcement and matrix challenging.

Fig. 12b and c show the expected P-F cohesive properties behave as two different parts divided by the turning points, marked as the circles. For the higher temperatures (210 and 220 ℃), the turning points are simply defined by the peak in the curve (i.e., when the gradient turns 0). For the lower temperatures (180 and 200 ℃), due to lack of obvious softening, the turning points are determined by finding the point where the second-order derivative of the stress–strain curve is approximately zero. For each cohesive stress curve, the rapid increase occurs before the turning point, while the cohesive stress either stops increasing rapidly or starts decreasing after the turning point. The turning points in the cohesive stress-composite strain curve indicates the beginning of the P-F interface debonding. As can be inferred, the cohesive behaviour shows temperature dependence. With the increase of temperature, there is a noticeable reduction in the stress level and the strain value at the turning point.

Based on the above analysis, a rheology model was proposed for the P-F interface part comprising a tensile spring, a frictional element and a pre-tensioned spring arranged as shown in Fig. 10. The frictional element is in parallel with the pre-tensioned spring and the two are in series with the tensile spring. At the initial stage of the deformation, only the tensile spring is activated, causing the stress to increase with the deformation. Once the stress in the frictional element reaches a certain value, the frictional element is activated and moves in conjunction with the pre-tensioned spring. According to the rheology form, it can be conjectured that with the further deformation, the tension in the pre-tensioned spring is reduced, causing a decrease in the total stress in this P-F interface part. Consequently, the stress in this P-F interface part changes. Depending on the stiffnesses of the tensile spring and

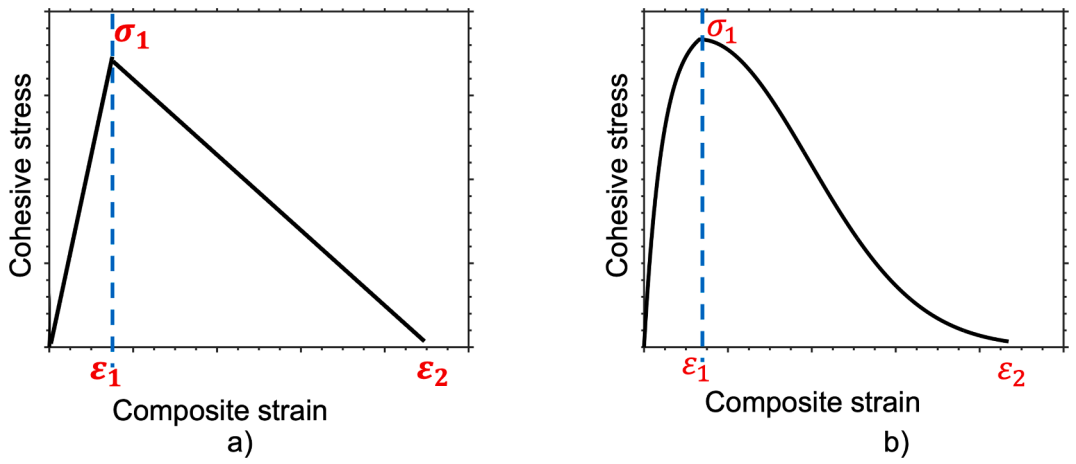


Fig. 13. a) the bilinear cohesive model; b) The exponential cohesive model proposed in this paper.

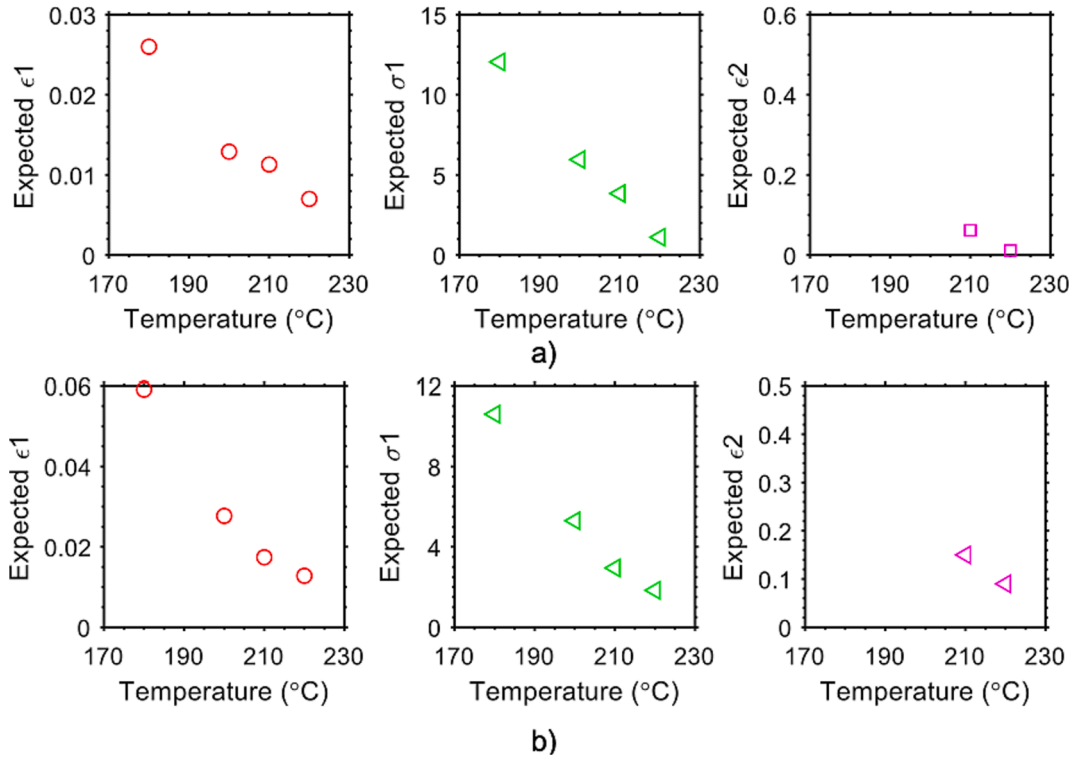


Fig. 14. The expected controlling parameters for cohesive stress curves under a) transverse tension and b) longitudinal shear deformation.

compressive spring, the stress may continue to increase, remain constant or begin to decrease. Therefore, the proposed rheology model can effectively represent the cohesive stress characteristics of the P-F interface. The constitutive model can be represented as Eqs. (12) and (13).

$$\boldsymbol{\sigma}_{P-F} = \begin{cases} \boldsymbol{\sigma}_{ts} \text{(before the friction element is activated)} \\ \boldsymbol{\sigma}_{ts} = \boldsymbol{\sigma}_{pre-ts} \text{(after the friction element is activated)} \end{cases} \tag{12}$$

$$\boldsymbol{F}_{P-F} = \begin{cases} \boldsymbol{F}_{ts} \text{(before the friction element is activated)} \\ \boldsymbol{F}_{ts} \cdot \boldsymbol{F}_{pre-ts} \text{(after the friction element is activated)} \end{cases} \tag{13}$$

where $\boldsymbol{\sigma}_{P-F}$ and $\boldsymbol{F}_{P-F}$ refer to the Cauchy (true) stress tensor and deformation gradient tensor in the P-F. Meanwhile, $\boldsymbol{\sigma}_{ts}$, $\boldsymbol{\sigma}_{pre-ts}$ are the stress tensors in the tensile spring and pre-tensioned spring, respectively, and $\boldsymbol{F}_{ts}, \boldsymbol{F}_{pre-ts}$ are the deformation gradient tensors in the tensile spring and pre-tensioned spring, respectively.

As for the cohesive model, a bilinear cohesive model is commonly used in microscale studies such as the RVE studies [54,55] and it is illustrated in Fig. 13a. The bilinear cohesive model is also embedded in ABAQUS [55]. However, in the macroscale model, as indicated in Fig. 12b and c, a more comprehensive model form is required to better capture the characteristics. Therefore, an exponential model [56] was selected to represent the macroscale response of the cohesive behaviour, as schematically illustrated in Fig. 13b.

The controlling parameters for the exponential cohesive model are marked in Fig. 13b. Specifically:

- The term ε_1 is the strain when the cohesive stress comes to a turning point.
- The term σ_1 is the stress value corresponding to the turning point.
- The term ε_2 is the strain when the cohesive model curve approaches zero, signifying, the complete debonding strain of cohesive elements.

Table 3
Summary of the unified constitutive model.

Composite model (C)

$$\sigma_C(\varepsilon_C) = \sigma_{PrF}(\varepsilon_{PrF}) + \sigma_{P-F}(\varepsilon_C)$$

Contribution of polymer constituent reinforced by Fibres (PrF)

$$\varepsilon_{PrF} = k^{map} \cdot \varepsilon_C$$

where k^{map} is the mapping relationship

For transverse tension: $k_{\varepsilon_{22}}^{map} = \bar{\varepsilon}_{equ}^{polymer} \cdot V_{F_p}/\varepsilon_{22}^{RVE}$

For longitudinal shear: $k_{\varepsilon_{12}}^{map} = \bar{\varepsilon}_{equ}^{polymer} \cdot V_{F_p}/\varepsilon_{12}^{RVE}$

Constitutive model of polymers, adapted from [40]:

$$\sigma_{PrF} = (\sigma_I + \sigma_N) \cdot Cor(\theta)$$

- The intermolecular resistance (I)

$$\sigma_{lim} = \sigma_{ref}(1 - ((\theta - \theta_{ref})/(\theta_m - \theta_{ref}))^{m_\sigma})$$

- Viscoelastic part ($\sigma_I < \sigma_{lim}$)

$$\sigma_I = 2 \cdot C_1(\theta) \cdot \left(\lambda_{Ie}^2 - \frac{1}{\lambda_{Ie}}\right);\ \lambda_{Ie} = \exp(\varepsilon_{Ie})$$

$$C_1(\theta) = (E_{ref} + E_1 \cdot (\theta - \theta_{ref})^{m_E})/(4 \cdot (1 + \nu))$$

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon}_{PrF} - \dot{\varepsilon}_{Iv};\ \dot{\varepsilon}_{Iv} = \dot{\gamma}_{01} \cdot \left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right)^{m_{rI}} \cdot \left(\frac{\sigma_I}{\eta_{Iv}(\theta)}\right)^{m_{Iv}}$$

$$\eta_{Iv}(\theta) = \eta_{Iv0}/\left(\frac{\theta - \theta_g}{\theta_m}\right)^{m_{\theta I}}$$

- Viscoplastic part ($\sigma_I \geq \sigma_{lim}$)

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon}_{PrF} - \dot{\varepsilon}_{Iv} - \dot{\varepsilon}_{Ip};\ C_p(\theta) = C_{p1}/\left(\frac{\theta}{\theta_m}\right)^{m_{cp}}$$

$$\dot{\varepsilon}_{Ip} = \dot{\varepsilon}_{0p} \cdot \left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right)^{m_{Ip}} \cdot \left\{\exp\left(\frac{1}{C_p(\theta)} \cdot \left(\frac{\sigma_I}{\sigma_{lim}} - 1\right)\right) - 1\right\}$$

- Network resistance (*N*)

$$\sigma_N = \frac{C_R(\theta)}{3J_{Ne}} \cdot \frac{\bar{\lambda}_L}{\bar{\lambda}_{Ne}} \cdot \mathcal{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) \cdot \left(\frac{2}{3} \cdot \left((\bar{\lambda}_{Ne})^2 - 1/\bar{\lambda}_{Ne}\right)\right)$$

$$\bar{\lambda}_{Ne} = \sqrt{(\exp(2 \cdot \varepsilon_{Ne}) + 2\exp(-\varepsilon_{Ne}))/3}$$

$$\mathcal{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) = 3 \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) + \frac{9}{5} \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^3 + \frac{297}{175} \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^5 + \frac{1539}{875} \cdot \left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^7$$

$$C_R(\theta) = C_{R_{ref}} + C_{r1} \cdot (\theta - \theta_{ref});\ \eta_{Nv}(\theta) = \frac{\eta_{Nv0}}{\left(\frac{\theta - \theta_g}{\theta_m}\right)^{m_{\theta N}}}$$

$$\dot{\varepsilon}_{Ne} = \dot{\varepsilon}_{PrF} - \dot{\varepsilon}_{Nv};\ \dot{\varepsilon}_{Nv} = \dot{\gamma}_{02} \cdot \left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right)^{m_{rN}} \cdot \left(\frac{\sigma_n}{\eta_{Nv}(\theta)}\right)^{m_{Nv}}$$

- Temperature evolution and thermal softening

$$Cor(\theta) = \left(1 - \frac{\theta_{cor} - \theta}{\theta_m - \theta}\right)^{m_h};\ \theta_{cor} = \theta + \int \dot{\theta}$$

$$\dot{\theta} = \frac{\omega(\dot{\varepsilon})}{\bar{C}(\theta)}\left(\sigma_{Iv} \cdot \dot{\varepsilon}_{Iv} + \sigma_{Ip} \cdot \dot{\varepsilon}_{Ip} + \sigma_{Nv} \cdot \dot{\varepsilon}_{Nv}\right)$$

$$\omega(\dot{\varepsilon}_{PrF}) = \begin{cases} 0 & ;\ if\ \dot{\varepsilon}_{PrF} \leq 0.01 \\ \min(0.33 \cdot (\log_{10}\left(\frac{\dot{\varepsilon}_{PrF}}{\dot{\varepsilon}_{ref}}\right), 1) & ;\ if\ \dot{\varepsilon}_{PrF} > 0.01 \end{cases}$$

$$\bar{C}(\theta) = \bar{C}_{ref} + \bar{C}_1 \cdot (\theta - \theta_{ref})$$

Contribution of Polymer-fibre interface (P-F)

$$\sigma_{P-F}(\varepsilon_C) = \begin{cases} \left(-e^{\left(-\frac{3}{\varepsilon_1} \cdot \varepsilon_C + 10\right)} + e^{10}\right) \cdot \frac{\sigma_1}{(e^{10} - e^7)} & (\varepsilon_C \leq \varepsilon_1) \\ \sigma_1 \cdot e^{-\left(\frac{\varepsilon_C - \varepsilon_1}{\varepsilon_2 - \varepsilon_1} \cdot 2\right)^2} & (\varepsilon_C > \varepsilon_1) \end{cases}$$

$$\varepsilon_1 = \varepsilon_{10} \cdot e^{k_{\varepsilon 1} \cdot (\theta - \theta_m)}$$

$$\sigma_1 = \sigma_{10} + k_{\sigma 1} \cdot (\theta - \theta_m)$$

$$\varepsilon_2 = \varepsilon_{20} \cdot e^{k_{\varepsilon 2} \cdot (\theta - \theta_m)} + b_{\varepsilon 2}$$

Table 4
Summary of all the material constants in the unified constitutive model.

Polymer constituent (PA6) reinforced by fibres (PrF) (material constants from reference [40])										
General properties	ν	$\overline{C}_{ref}$(kJ/m^{3}K)	$\overline{C}_1$	θ_{ref}(K)	θ_g(K)	θ_m(K)	m_h			
	0.48	3.00	0.01	296	320	503	0.5			
Viscoelastic properties	E_{ref}(MPa)	E_1(MPa)	m_E	σ_{ref}(MPa)	m_σ	$\dot{\gamma}_{01}$	m_{rI}	m_{Iv}	η_{Iv0}	$m_{\theta I}$
	2400	−1041.2	0.1554	12.0	3.257	0.001	0.8	0.5	0.25	1.20
Viscoplastic properties	$\dot{\varepsilon_{0p}}$	m_{Ip}	C_{p1}	m_{cp}						
	0.0010	0.5	0.05	3.0						
Network properties	$\dot{\gamma_{02}}$	m_{rN}	m_{Nv}	η_{Nv0}	$m_{\theta N}$	$\overline{\lambda}_L$	$C_{R_{ref}}$	C_{r1}		
	0.001	0.1	0.10	490	0.01	2.60	23.958	−0.120		
Polymer contribution mapping parameter K_{map}										
Transverse tension:$k^{map}_{\varepsilon_{22}}$ = 1.76					Longitudinal shear:$k^{map}_{\varepsilon_{12}}$ = 1.19					

Polymer-fibre interface (P-F)							
	ε_{10}	$k_{\varepsilon 1}$	σ_{10}(MPa)	$k_{\sigma 1}$(MPa/K)	ε_{20}	$k_{\varepsilon 2}$	$b_{\varepsilon 2}$
Transverse tension	0.0045	−0.0328	−2.542	−0.272	6.0 x 10^{-8}	−0.55	0.011
Longitudinal shear	0.0075	−0.0390	−1.710	−0.226	7.0 x 10^{-8}	−0.55	0.089

These controlling parameters, ε_1, σ_1 and ε_2 are temperature-dependent parameters. Based on the expected cohesive stress curves shown in Fig. 12b and c, the expected values of these controlling parameters, namely expected ε_1, expected σ_1 and expected ε_2 can be extracted. These values are shown in Fig. 14.

Note that for expected ε_2, Fig. 12b and c show that there is no obvious failure strain under lower temperatures (180 and 200 ℃), so it will be considered as infinity (∞) in this paper and represented by a relatively big number in the later fitting process. In Fig. 14, only the values at 210 and 220 ℃ are shown.

The two-part cohesive behaviour shown in Fig. 13b can be represented by a piecewise-defined function: the first part is before reaching turning point, where the cohesive stress continuously increases with the increase of strain; the second part is after the turning point. The piecewise-defined function form is proposed as Eq. (14).

$$\sigma_{P-F}(\varepsilon_C) = \begin{cases} \left(-e^{\left(-\frac{3}{\varepsilon_1}\cdot\varepsilon_C+10\right)}+e^{10}\right)\cdot\sigma_1/(e^{10}-e^{7})(\varepsilon_C \le \varepsilon_1) \\ \sigma_1\cdot e^{-((\varepsilon_C-\varepsilon_1)/(\varepsilon_2-\varepsilon_1)\cdot 2)^2}(\varepsilon_C > \varepsilon_1) \end{cases} \quad (14)$$

where, the two formulas calculate the cohesive stresses of the first and second parts of the cohesive model, respectively. The constant values e^{10} and e^{7} are directly selected to represent considerable large numbers for obtaining the desired stress level. Other considerations are: for the continuity of the piecewise-defined function, $\sigma_{P-F}(\varepsilon_1) = \sigma_1$; $\sigma_{P-F}(\varepsilon_2)$ nears zero; marking the first-order derivative of the function as $\partial\sigma_{P-F}(\varepsilon_C)/\partial\varepsilon_C$, then $\partial\sigma_{P-F}(\varepsilon)/\partial\varepsilon$ is considerably large when $\varepsilon_C = 0$, and nears zero when $\varepsilon_C = \varepsilon_1$ and ε_2.

When it comes to the relationship between the controlling parameters, ε_1, σ_1 and ε_2, and temperatures, it can be expressed as Eqs. (15)-(17) according to Fig. 14.

$$\varepsilon_1 = \varepsilon_{10}\cdot e^{k_{\varepsilon 1}\cdot(\theta-\theta_m)} \quad (15)$$

$$\sigma_1 = \sigma_{10} + k_{\sigma 1}\cdot(\theta-\theta_m) \quad (16)$$

$$\varepsilon_2 = \varepsilon_{20}\cdot e^{k_{\varepsilon 2}\cdot(\theta-\theta_m)} + b_{\varepsilon 2} \quad (17)$$

where, $\varepsilon_{10}, k_{\varepsilon 1}, \sigma_{10}, k_{\sigma 1}, \varepsilon_{20}, k_{\varepsilon 2}$ and $b_{\varepsilon 2}$ are all cohesive-related material constants, and θ and θ_m refer to the current temperature and the melting temperature of the polymer matrix, respectively. Note that all temperatures were converted to absolute temperatures in Kelvin (K) in the constitutive modelling.

By fitting the Eqs.(14)-(17) to Fig. 12b and c, the cohesive-related constants are determined. The material constants together with the whole model are listed in Table 4.

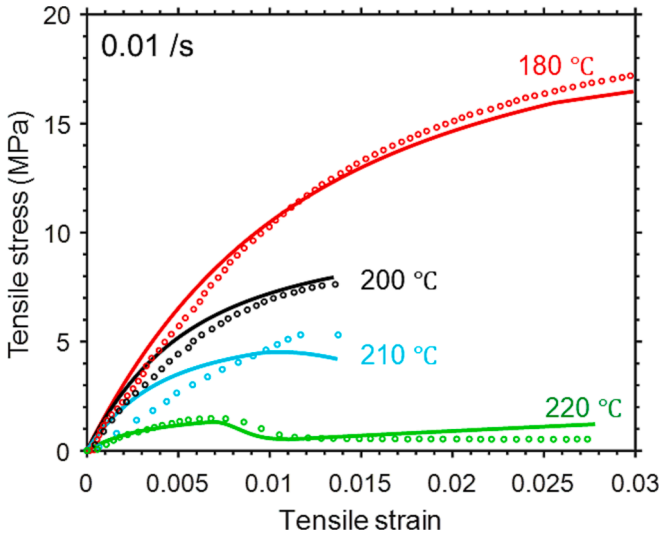


Fig. 15. The comparison between the constitutive model predictions (solid lines) and the experimental results (circles) of transverse tensile tests.

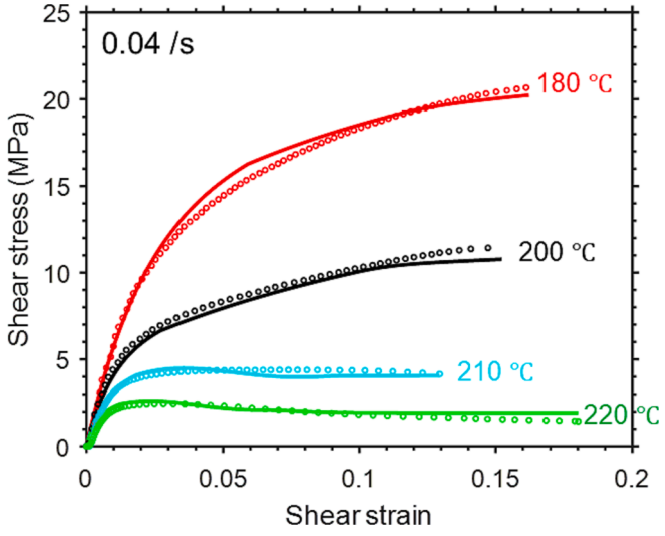


Fig. 16. The comparison between the constitutive model predictions (solid lines) and the experimental results (circles) of VNR shear tests.

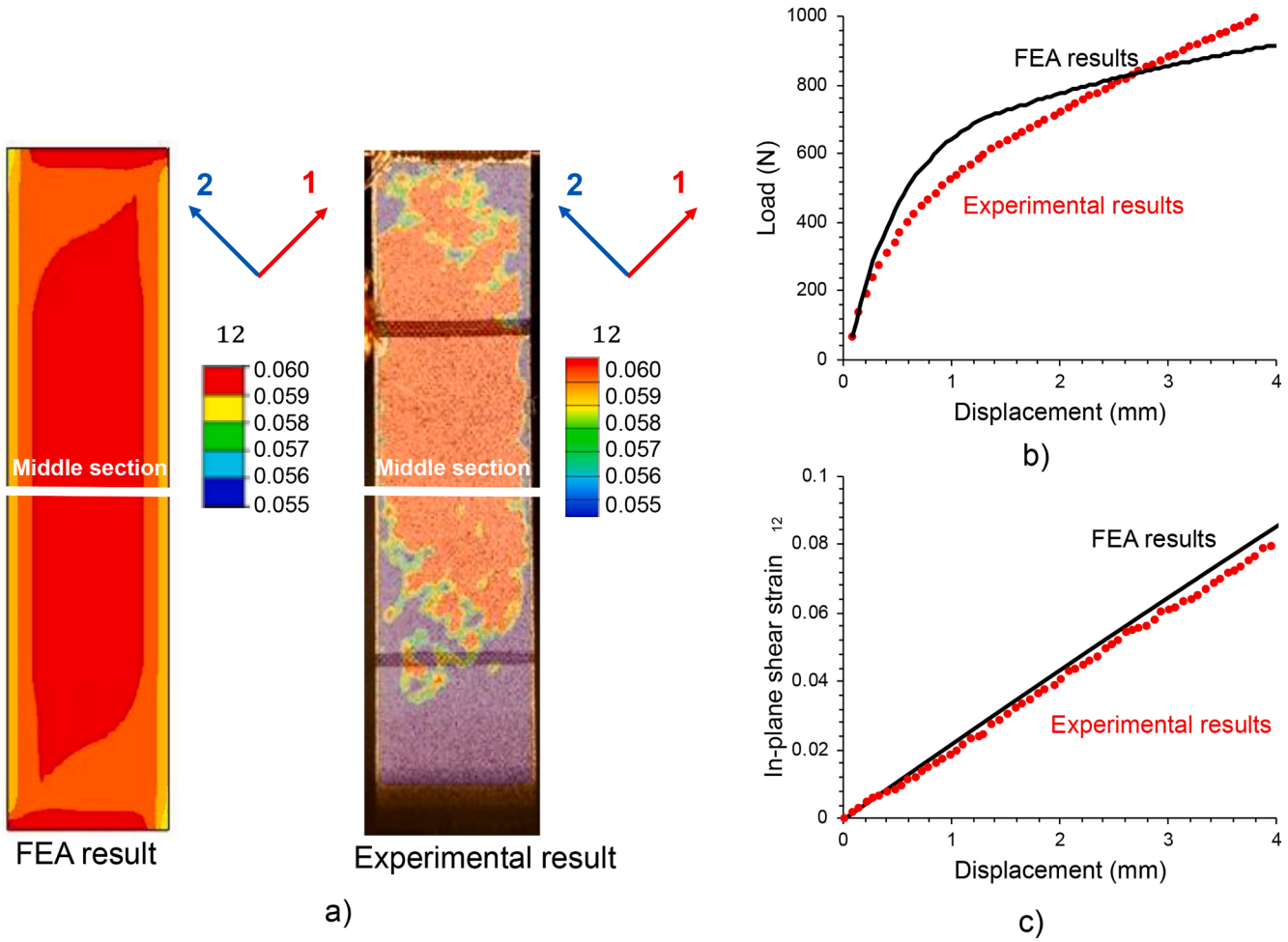


Fig. 17. The comparison between the FEA and experimental results of a bias-extension test. a) Comparison of the in-plane shear strain distribution at the displacement of 2.7 mm (1 is the fibre direction); b) The comparison of the loads measured experimentally and numerically; c) The comparison of the history of the averaged in-plane shear strain across the middle section of the specimen.

4.3. The constitutive model and its performance

The entire constitutive model and all model constants are summarised in Table 3 and Table 4.

Fig. 15 and Fig. 16 show the model prediction compared to experimental results for transverse tensile and VNR shear tests, respectively. The results show that the model can predict the transverse tensile and longitudinal shear properties with satisfying accuracy. The temperature effects are well reflected, and the strain-hardening or -softening features are well captured. The model takes the intra-ply tensile and shear deformation mechanisms into account, therefore, can be used later for numerical simulation of the solid-state stamping of sheet UD FRTPs.

5. Implementation and validation of the developed constitutive model

The developed constitutive model was implemented into ABAQUS/Explicit using a user defined subroutine and subsequently used to simulate a bias-extension deformation. The validity and accuracy of this constitutive model can be evaluated by comparing the FEA and experimental results of the bias-extension deformation.

The experimental setup of the bias-extension test is the same as the experimental setup of the transverse tension test. A bias-extension specimen with a dimension of 15 x 150 x 3.3 mm and a freely deforming length of 56 mm was tested. The configuration of the composite laminates was $[+45_3/-45_3]_{2s}$. The bias-extension test was conducted at 200 ℃ with a crosshead speed of 50 mm/min. The strain field of the bias-extension specimen was also processed via the DIC method in GOM software with the same setting as stated in Section 2.3.

To investigate the accuracy of the constitutive model, a macroscale FEA model was constructed with the same dimensions and carbon fibre configurations as the experimental specimen. The specimen at a thickness of 1.65 mm - half the thickness of the experimental specimen, was partitioned into four sections in the thickness direction, with material orientation of $[+45/-45]_2$. For the boundary conditions for the specimen, a vertical displacement of 4 mm was applied to the top edge, while the bottom edge was fixed, and the back surface was set to be the symmetrical surface. The solid element (C3D8R) was selected for the specimen.

The shear strain field (ε_{12}) obtained from the FEA and experimental results at a displacement of 2.7 mm are compared in Fig. 17a. It is evident that the simulation using the developed constitutive model accurately predicts the shear strain field of the bias-extension deformation. The experimentally measured and FEA calculated load–displacement curves were compared, and a reasonably good agreement was achieved, as shown in Fig. 17b. Additionally, the averaged shear strains of the middle section marked by the white lines in Fig. 17a were compared and presented in Fig. 17b. The FEA results were calculated as the average value of all element values of the middle section, while the experimental results were computed as the average value of all the facets crossed by the middle section. Fig. 17 demonstrates that the simulation using the developed constitutive model can

effectively predict the in-plane deformation of UD CF/PA6.

6. Conclusions

This paper presents the experimental and constitutive modelling studies on the thermomechanical properties of unidirectional fibre-reinforced thermoplastics (UD FRTPs) under solid-state stamp forming conditions. Unidirectional carbon fibre-reinforced polyamide 6 (UD CF/PA6) was used as the studied material. The intra-ply transverse tensile and longitudinal shear properties of UD CF/PA6 have been systematically investigated to gain insights into the intra-ply deformation mechanisms. The insights gained from the experimental characterisation and analysis have informed the development of a new unified constitutive model. The main conclusions are drawn as follows:

1) The temperature has significant effects on the intra-ply properties of UD CF/PA6 under the investigated conditions and deformation modes (180 – 220 ℃, at the strain rate of 0.01/s for the transverse tensile and 0.04/s for the longitudinal shear tests). In the studied temperature range, strain-hardening behaviour occurs after the initial elastic deformation at all temperatures; whereas at the higher temperatures (i.e., 210 and 220 ℃), the strain-hardening gradually disappears and the strain-softening occurs.
2) The strain-rate effects on the intra-ply deformation properties of UD CF/PA6 are marginal under the studied conditions and deformation modes (0.01 – 0.25 /s for the transverse tensile and 0.004 – 0.4 /s for the longitudinal shear tests, both are at 200 ℃).
3) The performance of the new constitutive model is demonstrated by the good agreements between the experimentally measured stress–-strain curves and those predicted by the model under transverse tensile and longitudinal shear deformations, taking into account temperature effects.
4) The effectiveness of the implementation of the developed constitutive model into FEA commercial software has been confirmed through the bias-extension simulation using ABAQUS/Explicit with a user defined subroutine. The good agreements between the numerical and experimental results further validate the performance of the new constitutive model and its potential in process simulations for predicting the intra-ply deformation in UD CF/PA6 under solid-state stamp forming conditions.

In summary, the methodologies and model developed in this study represent a novel approach to characterise and model UD FRTPs. This approach can be readily applied to other UD FRTPs, serving as a versatile tool for future research.

CRediT authorship contribution statement

Hongyan Wang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Zerong Ding:** Formal analysis, Methodology, Writing – review & editing. **Xingtong Chen:** Data curation, Formal analysis. **Haibao Liu:** Methodology, Supervision, Writing – review & editing. **Nan Li:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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